



Generative adversarial networks (GANs): Introduction, Taxonomy, Variants, Limitations, and Applications

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Abstract

The growing demand for applications based on Generative Adversarial Networks (GANs) has prompted substantial study and analysis in a variety of fields. GAN models have applications in NLP, architectural design, text-to-image, image-to-image, 3D object production, audio-to-image, and prediction. This technique is an important tool for both production and prediction, notably in identifying falsely created pictures, particularly in the context of face forgeries, to ensure visual integrity and security. GANs are critical in determining visual credibility in social media by identifying and assessing forgeries. As the field progresses, a variety of GAN variations arise, along with the development of diverse assessment techniques for assessing model efficacy and scope. The article provides a complete and exhaustive overview of the most recent advances in GAN model designs, the efficacy and breadth of GAN variations, GAN limits and potential solutions, and the blooming ecosystem of upcoming GAN tool domains. Additionally, it investigates key measures like as Inception Score (IS) and Fréchet Inception Distance (FID) as critical benchmarks for improving GAN performance in contrast to existing approaches.

Keywords Digital forensics · Image vision · Deep learning · Generative adversarial network · Deep fakes · GAN models · Deep learning based methods · GAN architecture · GAN variants

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1 Introduction

"Generative adversarial learning-based approach to generative modeling that incorporates deep learning -based strategies like CNN (convolutional neural networks) referred to as a GAN. In the discipline of deep learning, generative modeling is a sort of unsupervised learning assignment that involves detecting and acquiring knowledge of the general patterns or structures in input data thus that the model can be utilized to construct or output new instances that might have been selected from the entire data. GAN versions are now used to produce images from text data, videos from still visuals, increase image resolution, or alter images. Its applications range from anomaly spotting to chess game improvements [1].

Recent advancements in generative adversarial networks (GANs) create human facial photographs that are difficult for people to identify with their naked eyes, known as "Deepfake". Perverting these photographs on the internet can lead to ethical, moral, and legal concerns. As new GANs emerge at a rapid pace, generalizing forensics models to detect new GAN pictures is an important research topic. The forensics community creates novel ways to detect fake images, particularly face forgeries, and protecting visual trustworthiness. There is no denying that the field of deep learning research is getting increasingly interested with GANs across a wide range of applications. The Basic GAN architecture is mentioned in Fig. 1.

The first GAN model was developed by Ian J. Good et al. [2] and it was released in 2014. The GANs are unsupervised generative models that automatically locate the data distribution [3]. The "Generator" and the "Discriminator" are the two main models in a GAN system. One type of neural network, a convolutional network, is responsible for generating data, while another type of network, a deconvolution network, is used for classification. Conceptually, GAN uses a supervised learning approach to generate synthetic data, allowing it to engage in unsupervised learning. The model assumes the structures present in data sets without taking into consideration findings that have been anticipated, labelled, or classified [4]. By introducing adversarial learning between the generator and the discriminator, GAN avoids the trap of maximizing likelihood. D works to maximize the likelihood that it correctly distinguishes between reals and fakes by increasing the value of $\log D(x)$, while G strives to minimize the likelihood that D will predict that its outputs are fake by minimizing the value of $\log (1 - D(G(z)))$. This is a minimax game, as stated by Good fellow. The GAN formula can be found in equation 1, which also contains its definition.

$$GminDmaxV(D, G) = Ex \sim Pdata(x)[loadD(x)] + Ez \sim pz(z)[\log(1 - D(G(z)))] \quad (1)$$

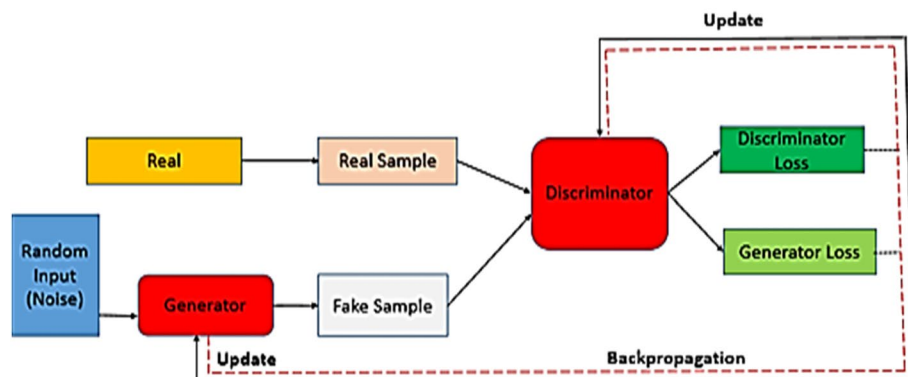


Fig. 1 Basic GAN architecture

Equation 1: Mathematical formula of GAN [2]

Where; G stands for Generator, D stands for Discriminator, $P_{data}(x)$ = real-world data distribution, $P(z)$ = generator distribution, $x = P_{data}$ sample (x), z = a sample taken from $P(z)$, $D(x)$ denotes a discriminator network, $G(z)$ denotes the generator network. The basic working steps of GAN model is stated and shown on basis of the flow chart in Fig. 2.

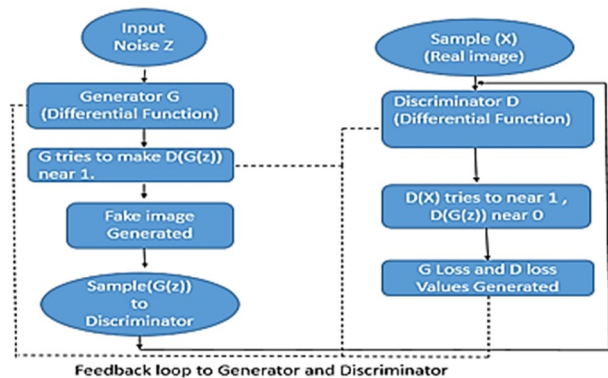
1.1 GAN Execution Steps:

1. A random number generator returns an image.
2. The discriminator handles both pictures from the dataset and the generator's output.
3. Real and fake images are given to the discriminator, which returns probabilities between 0 and 1
4. The discriminator has ground truth images from the training dataset based on discriminator losses.
5. The generator and discriminator are in a feedback loop to reduce generator losses.

“DeepFakes” [5] as mentioned is a well-known application of GANs that aids seamless transformation of one person's face into another's in both photos and videos. With the continuous improvement of GAN capabilities, the creation of convincing yet deceptive images and videos has become increasingly easy, posing challenges for human observers in distinguishing their authenticity. Figure 3 shows generation of fake images from real images using Deepfake technology. The generated faces do not hold any processing information because GAN generates images using a random vector. Even with better computer tools, it's getting more and more difficult to locate these kinds of photos. Concerns about facial modification, including the synthesis of full facial pictures, are at a high point of concern in field of machine and deep learning nowadays. Concerns regarding potential for abuse are understandable because large-scale face databases, along with significant improvements in deep learning technology, have allowed for the generation of incredibly convincing fake facial material. There are moral, societal, and security concerns associated with the widespread distribution of fraudulent online identities and content (e.g., fake news, fraud). Therefore, people place a higher importance on the veracity of visuals. As a result of these concerns, research into forgery detection methods has sprung up [6].

GANs have developed into a variety of models with a range of performance potentials [7]. There are several distinct types of generative adversarial networks, and each one operates in a little slightly different way and helps researchers get slightly different

Fig. 2 GAN execution flowchart



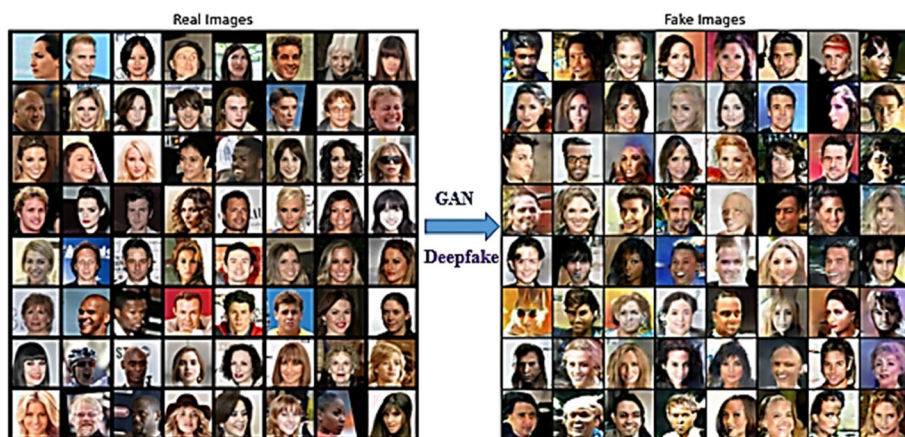


Fig. 3 Generation of fake images from real images using Deepfake

results. GAN's architectures are positively impacting its prominence and have shown excellent results in modifying visual styles. Table 1 defines as abbreviation table used for studies of different GAN variants discussed in the paper detailing about their respective referencing and author.

Contribution:

- 1) Comprehensive Review of GAN Model Architectures highlighting the progress of innovations from the introduction of GANs in 2015 to the changes seen till 2022 by providing important insights into the changing environment of generative models.
- 2) An in-depth investigation into GAN efficiency metrics, addressing training issues and improving performance in image diversity, image quality, and training stability.
- 3) Examining important study findings, the paper highlights future research goals and areas for development in analyzing GAN variants, improving synthetic information generation, and addressing face forgeries.

This review study tries to investigate and analyze the modifications to GAN architecture by the creation of GAN variants discussed in the review of literature of previous eight years (2016-2022) in section 2. Section 3 defines various GAN variants in detail used in generating various architecture of GAN by adding new layers. The limitations related to GAN techniques such as training difficulties, data handling issues, system instability, false predictions, etc. are thoroughly detailed with optimal solution and discussed in section 4 and section 5. Various applications for GANs including image editing and synthesis, image translations as well as audio enhancement and synthesis using the reserves from the GAN models are discussed in section 6. Section 7 identifies different evaluation parameters like FID and IS as major parameters used for evaluation of GAN models that help to improve its performance. Various types of available datasets are discussed in section 8. Section 9 discusses the various research findings and discussion about the review. Section 10 finally conclude this review study.

Table 1 Abbreviation table of different GAN variants used in the paper

<i>GAN Variant</i>	<i>Author</i>	<i>Reference</i>
DCGAN	Radford et al	[20]
Pix2Pix	Isola et al	[21]
StackGAN	Zhang et al	[22]
SRGAN	Ledig et al	[23]
CoGAN	Liu et al	[24]
Progressive Growing GAN	Karras et al	[25]
Cycle GAN	Zhu et al	[26]
TAC-GAN	Dash et al	[27]
GP-GAN	Wu et al	[28]
WGAN	Arjovsky et al	[29]
AttnGAN	Xu et al	[30]
SN-GAN	Miyano et al	[31]
ESRGAN	Wang et al	[32]
AlphaGAN	Lutz et al	[33]
Misc-GAN	Zhou et al	[34]
SAGAN	Zhang et al	[35]
DVDGAN	Clark et al	[36]
BIGGAN	Brock et al	[37]
LocalGAN	Daras et al	[38]
StyleGAN	Cheong et al	[39]
Self Supervised GAN	Tran et al	[40]
SARAGAN	Yuan et al	[41]
FAGAN	Jiang et al	[42]
OodGAN	Marek et al	[43]
PiGAN	Chen et al	[44]
SSGAN	Bi et al	[45]
OurGAN	Yoom et al	[46]
TGGAN-AD	Xu et al	[47]
Sphere GAN	Park et al	[48]
WGAN-GP	Gulrajani et al	[70]
LAPGAN	Denton et al	[71]
Triple GAN	Li et al	[73]
Disco GAN	Kim et al	[74]
FA-GAN	Chang et al	[78]
Multi-GAN	Hatua et al	[81]
TTS-GAN	Li et al	[86]
C-RNN GAN	Mogren et al	[94]
ORGAN	Guimaraes et al	[95]
SeqGAN	Yu et al	[96]
ChemGAN	Behenda et al	[98]
MedGAN	Choi et al	[101]

Fig. 4 Categorization of Density Function as Implicit Function and Explicit Function

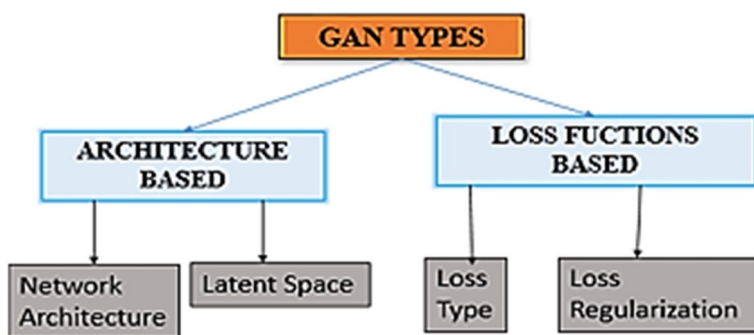
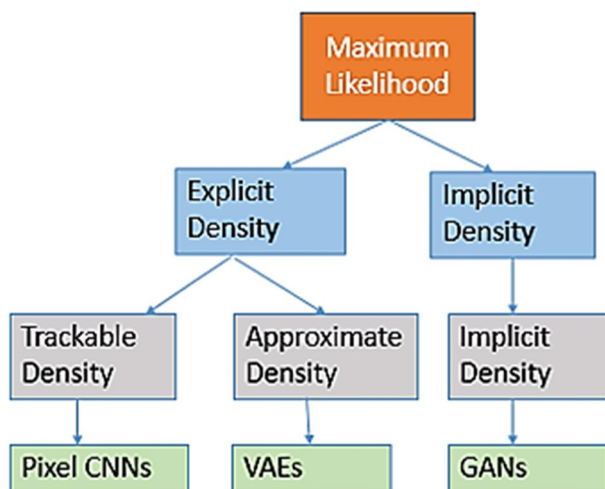


Fig. 5 Categorization of GAN

2 Review of literature

This section primarily consists of discussion about the various review studies of existing popular GAN models based on their different architecture and loss optimization from year 2015-2022 in section 2.1. Then, in section 2.2 the taxonomy of the literature of existing GAN of last eight years that is based on variances in GAN structures and loss functions. Also, the role of CNN in developments of cutting-edge technology of GAN models, GAN variants and their application scopes are significantly reviewed in section 2.3 to provide critical elaboration and insights of the topic. This study categorizes Density Function as Implicit Function and Explicit Function as shown in Fig. 4. Also, classifies the current generation of GANs into two primary variations, which we refer to as the architecture version and the loss variant. The classification of few popular GANs on basis of architectures and Loss function is depicted in Fig. 5.

2.1 Generative models: Exploring relationships and application environment

Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Flow-Based Models, and Diffusion Models are all different types of generative models, each with their own set of strengths and weaknesses. Due to their complimentary nature, GANs and VAEs have been widely investigated together. GANs use adversarial training to encourage realistic sample production, whereas VAEs provide a probabilistic framework for encoding and decoding latent representations. Notable publications, such as "Auto-Encoding Variational Bayes" by Kingma and Welling (2013) and "Generative Adversarial Nets" by Goodfellow et al. (2014), have helped to create linkages between these two fundamental techniques. Furthermore, the relationship between GANs and flow-based models is explored in works such as "Density Estimation using Real NVP" by Dinh et al. (2017), which examines the potential synergy between Gans' implicit generative capabilities and flow-based models' explicit density estimation methods. The intersection with Diffusion Models, as proven by Gulrajani et al. (2017)'s "Improved Training of Wasserstein GANs," adds to our understanding of generative modeling by including the diffusion process as an alternative to conventional sampling strategies. Investigating these links gives a full perspective of the generative modeling landscape and paves the way for hybrid models that combine the characteristics of each technique to improve overall performance and reliability.

The adversarial training strategy allows GANs to capture complicated data distributions while generating high-fidelity outputs. VAEs, on the other hand, are appropriate for applications that need ordered latent representations, such as image reconstruction and generation with specified attributes. The probabilistic encoding-decoding method enables precise control over the generation process. Flow-based models are suitable for situations requiring explicit density estimates and invertible mappings, such as uncertainty quantification and anomaly identification. Diffusion models are ideal for large-scale image generation applications that need effective training. GANs, VAEs, Flow-Based Models, and Diffusion Models have the following applications:

- Generative adversarial networks (GAN)
Applications: GANs are commonly used in picture creation, style transfer, and data enhancement. They excel at jobs that require the creation of realistic and high-fidelity samples, such as photorealistic photos, art, and even deepfakes.
- Variational autoencoders (VAEs)
Applications: VAEs excel in applications requiring organized latent representations, such as image reconstruction, picture production with controlled properties, and feature learning. They are especially effective in situations when explicit control of the creative process is required.
- Flow-based models
Applications: Flow-based models, such as Real NVP, are useful in situations that require explicit density estimates and invertible mappings. They are frequently used in activities like uncertainty quantification and anomaly identification.
- Diffusion models
Diffusion models, as investigated in "Improved Training of Wasserstein GANs," are useful in situations where steady and effective training is required. They are especially beneficial for large-scale picture production applications, improving training stability and convergence.

In a nutshell, GANs excel in tasks that require realistic sample generation, VAEs excel in tasks that require structured latent representations, flow-based models are useful for explicit density estimation and invertible mappings, and diffusion models improve training stability in large-scale generative tasks. The specific applications demonstrate the models' adaptability across several areas in the field of generative modeling.

2.2 Choosing GAN over other generative models and beyond

Generative Adversarial Networks (GANs) are preferable to other generative models due to their advantages under certain circumstances. GANs are particularly well-suited to applications that require the generation of extremely realistic and visually appealing samples. For example, GANs tend to outperform other models in creative areas such as art production, picture synthesis, and style transfer, where catching complex details and delivering lifelike outputs is crucial. The adversarial training process of GANs allows them to learn complicated data distributions, making them especially useful in cases where capturing the intricacies of real-world data is critical. Furthermore, GANs grow in applications that need diverse and innovative content production, such as the fashion sector for generating unique apparel trends or in entertainment for producing engaging computer-generated images.

While Generative Adversarial Networks (GANs) thrive in some contexts, other generative models are more appropriate and beneficial. Variational Autoencoders (VAEs), for example, are recommended when the goal is to learn a structured latent space and explicit control over the generating process is required. VAEs provide specific benefits in jobs requiring the production of data with interpretable and meaningful representations, such as controlled attribute synthesis in photos or feature learning in healthcare data.

Flow-Based Models, with their emphasis on explicit density estimates and invertible mappings, are the model of choice when a good grasp of the underlying data distribution is required. Flow-based models' tractable likelihoods make them useful for applications such as anomaly identification and uncertainty quantification. Furthermore, in circumstances requiring stable training and convergence, particularly in large-scale generating tasks, diffusion models outperform GANs by resolving training stability difficulties.

As a result, the choice between various generative models is determined by the task's unique criteria, such as the necessity for interpretable latent spaces, explicit density estimates, or improved training stability. Understanding the benefits and limits of each model type is critical for making educated decisions based on the data's properties and the generating task's aims.

2.3 Review studies of popular GAN models based on architecture and loss optimization

Several GAN variants are developed as enhancements of the base GAN model to eliminate its limitations of instability and complexities at the time of model training. The variants developed and are evaluated many times to identify the prime potentialities and scope of further improvements. The existing GAN variants find their importance in distinct areas that are frequently discussed in academic articles. Such endeavors are made to provide established resources of GAN's effectiveness to the growing application areas where GANs are being accepted and tried currently. This review too, has been worked to serve the same purpose and accordingly provides scrutiny of articles that talk

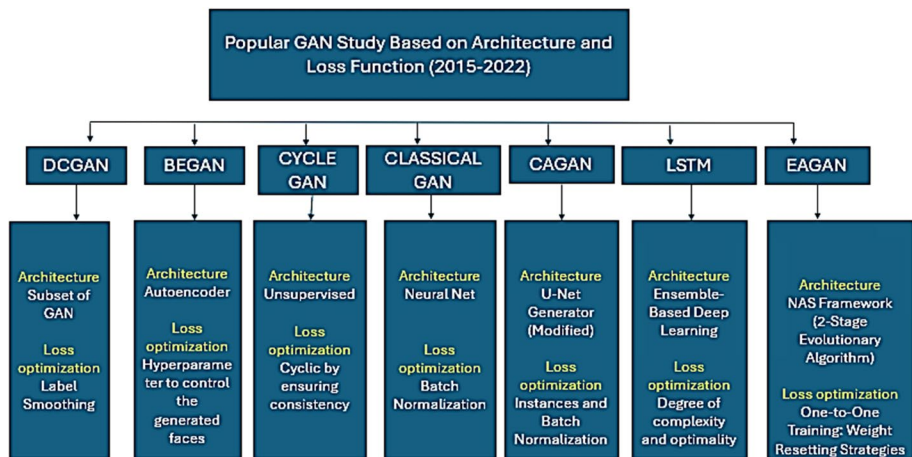


Fig. 6 Review studies of existing GAN models based on their different architecture and loss function optimization from year 2015–2022

about new and traditional GAN variants as shown in Fig. 6. Table 2 shows experimental details on the literatures reviewed on GAN Architecture (2015 - 2022).

The state-of-the-art GANs, which included base GAN, WGAN, Semi-GAN, C-GAN, LS-GAN, BiGAN, AC-GAN, InfoGAN, SeqGAN, and BEGAN, were reviewed and summarised by [15]. The proposal background, theoretical and practical models, and application domains of GAN variations were initially examined. Then, the benefits, drawbacks, and development patterns of GANs were analysed. The authors specifically looked at the connection connecting GANs and parallelism logic, concluding that GANs holds lot of caliber for virtual-real interaction and integration in parallel systems research. Their experiment demonstrated that GANs provide significant algorithmic assistance for parallel intelligence. Lala et al. [16] compared contemporary Generative Adversarial Network architectures—in special, AdaGAN, VEEGAN, Wasserstein GAN, and Unrolled GAN—on data sources with various distributions, including synthetic and real datasets, and with regard to a number of widely accepted indicators for calculating mode collapse. According to the study’s findings, Wassertstein GAN underperformed on the datasets, but AdaGAN consistently outperformed the other GANs on almost all of them. The test findings also revealed that since the measures did not produce reliable results, one metric was insufficient to quantify mode collapse for GANs.

Yoo et al. [17] noted that a diverse family of GANs with improved performance in each generation was produced as a result of different sorts of GANs being investigated from various perspectives. Their review concentrated on distinct GANs by identifying their common characteristics based on this working resource. Handalage et al. [18] reviewed “Deep Convolutional GAN (DCGAN), Laplacian GAN (LapGAN), Wasserstein GAN (WGAN), Information Maximizing GAN (InfoGAN), Conditional GAN (CGAN), Energy-based GAN (EBGAN), Boundary Equilibrium GAN (BEGAN), Progressive Growing GAN (PGGAN), Big GAN (BigGAN), and Style-based Generative Architecture” for GANs. Additionally, their fundamental developmental mechanism and goal-oriented functions were investigated. Second, the most popular GAN variations were presented, along with their advantages and disadvantages. Thirdly, a few of

Table 2 Experimental details on the literatures reviewed on GAN architecture (2015–2022)

Research	Year	Research Purpose	Parameters	Dataset	Research Results	Performance Score
Uzun et al. [8]	2020	1. Assessment of the performance of GAN as a tool for synthesizing the proposed based on plan drawing outputs 2. Utilized Andrea Palladio's standalone architectural and engineering strategies with the DCGAN paradigm	Epochs: 10,000 Batch size: 32 Optimizer: Adam Learning rate: 0.0002, beta1:0.5, and beta2: 0.999	125 architect Andrea Palladio's original home blueprints (sixteenth century) and 150 Palladian plans generated using Palladian grammar rules as defined by Stiny & Mitchell (1978) with GRAPE SGI software	(1) For 1st test, the DCGAN results were too chaotic to reliably assess DCGAN's ability to generate plans Poor data integrity in the original strategy led to inaccurate samples (2) DCGAN started generating reasonable plans after the 600th epoch, at the 10,000th epoch, the training procedure was terminated	Model evaluated: (1) Qualitative evaluation method, as fast scene categorization in which just 63 of the 2525 plans created were in compliance with the Palladian grammar rule. Space syntax analysis also provided similar results (2) Quantitative evaluation method as FID. Although DCGAN did not follow Palladian grammatical rules, it became better at simulating the dataset

Table 2 (continued)

Research	Year	Research Purpose	Parameters	Dataset	Research Results	Performance Score
Gao et al. [9]	2020	Survey on GAN performance about quality of generated images in terms of (1) producing excellence images, (2) Mode diversification of output, (3) state of the art GAN approaches to eliminate Vanishing gradient	Shown as generic mathematical expression while discussing loss function of GAN models (CGAN, WGAN, WGAN-GP, UGAN, LS-GAN, RealnessGAN)	CIFAR10, ImageNet, LSUN and CelebA	(1) WGAN, RGAN, and LSGAN closely resemble the first GAN loss (2) RGAN and WGAN could solve vanishing gradient problems, while LSGAN suffered with the problem (3) SN-GAN is the inaugural GAN that is successfully applied to ImageNet datasets (4) SAGAN and BigGAN used spectral standardization, and achieved satisfactory results (5) The generalisation capacities of SN-GAN and RGAN are stronger	Evaluation for Human annotator and evaluation metrics (with CelebA dataset) (1) Accuracy measure (Human) - DCGAN: 0.995 BEGAN: 0.824 PROGAN: 0.705 (2) DCGAN: I/IS: 0.44 MMD: 0.22 FID: 63.29 I/Neuroscore: 1.715 BEGAN: I/IS: 0.57 MMD: 0.29 FID: 83.38 I/Neuroscore: 1.479 PROGAN: I/IS: 0.42 MMD: 0.12 FID: 34.10 I/Neuroscore: 1.195

Table 2 (continued)

Research	Year	Research Purpose	Parameters	Dataset	Research Results	Performance Score
Aggarwal et al. [10]	2021	Review on GAN models and their applications	Not Mentioned	Neurological Tumors of the Brain (NTU-RGB-D), Human 3.6 M, Brain Tumor Dataset (For GANs used in medical application); 3D CAD Datasets, such as, Shape Net, 2D-to-3D deformable sketches, ANN_SIFTIM, CIFAR-10, CLEF-IP 2011 (For GANs used in 3D image generation)	(1) Common areas of application found to be Topics such as 3D modelling, healthcare, epidemics, photography, recognising faces, transferring textures, and managing traffic flow are also covered. (2) Use of GAN and DL are getting more possibilities (3) Supervised, unsupervised, and reinforcement learning are more in use	In clinical image generation, COVID-GAN model used to for Identifying Covid-19 utilizing Radiograph, successfully did image classification with 89% accuracy

Table 2 (continued)

Research	Year	Research Purpose	Parameters	Dataset	Research Results	Performance Score
Dubenskaya et al. [11]	2021	Observation of GAN model to improve performance of image generation by preprocessing the training dataset	Generator used random vector of 8192 (128×8×8) entries, four by four filters, a leaky ReLU function with alpha equal to 0.2 (activation function). Discriminator used Input 32×32 greyscale image, 2×2 stride (downsample) taking the learning rate into account, Adam's version of stochastic gradient descent is 0.0002, momentum:0.5; Size of the convolution filter is 4×4, with alpha = 0.2 used to activate the leaky ReLU function; Batch size: 128 images and 300 epochs	Public dataset	The model performance was observed by drawing a graph. More over half of the created occurrences had computed probabilities between 90 and 100%. In addition, the chance for 97% of created occurrences exceeded 50%	Approximately 3% of the produced gamma pictures, which were previously very likely to be identified as non-gamma events, are now distinguished as gamma events. The final score was 97%

Table 2 (continued)

Research	Year	Research Purpose	Parameters	Dataset	Research Results	Performance Score
Namboodiri et al. [12]	2021	Improve efficacy of online retail fashion store in terms of customer attention boosting through quality image generation	It employed an enhanced generator built on a redesigned U-Net generator design. For the generator and discriminator, respectively, a mix of instance and batch normalisation was created. Additionally, the activation functions of ReLu and Leaky ReLu were combined	Zalando	The research introduced a photorealistic virtual try-on solution while ensuring optimization and minimalistic use of resources as required by industrial standards	The model achieved a superior quality in shorter training duration of 13 h on a smaller dataset of 10,167 images
Yazdanian et al. [13]	2021	To forecast cloud workload time-series independently for each sub-band depending on complexity and volatility, a novel state-of-the-art ensemble GAN/LSTM deep learning architecture (E2LG) is suggested	Deployed 1D ConvNets including its discriminator and generate stackable LSTM blocks	Standard workload traces for the cloud	By significantly exceeding both conventional and cutting-edge deep learning methods, E2LG significantly improved the accuracy with which cloud workloads could be predicted	Boosted the average accuracy of predictions by at least 5% compared to the state-of-the-art methods
Ying et al. [14]	2022	An innovative two-stage evolutionary encryption method centred on NAS framework for investigating GANs (EAGAN) was devised to combat the problem of programming GANs which were inconsistent	7.14M parameters for generators; 0.91M parameters for discriminator	CIFAR-10	EAGAN was applied to the task of unconditional picture synthesis, and it was able to conclude the search on the CIFAR-10 dataset in 1.2 GPU days with high efficiency	Performance Evaluation: IS = 8.81 ± 0.10 , FID = 9.91 Performed better than NAS-GANs, that were run on the STL-10 dataset, and the results were (IS = 10.44 ± 0.087 , FID = 22.18)

the industries and uses where GANs have been successful were examined. In this study, they created a GAN model and assessed the advantages and disadvantages of it.

Sharma et al. [19] compared the insights of CycleGAN (Cycle-Consistent Adversarial Network), a popular GAN for image-to-image translation, with Attention GAN (“Attention-Guided Generative Adversarial Network”). Their initial model, called CycleGAN, used two generators, two discriminators, and did not require a dataset of paired images to convert an image from one domain to another. The second method was AttentionGAN, which combined concentration and content overlays with the output from one field to produce a very realistic image in a new domain. These two were trained on the “CelebA-HQ (CelebFaces Attributes high-quality dataset) and FFHQ datasets for comparison (Flickr Faces HQ)”. Effectiveness was assessed quantitatively using identity preservation and five criteria for evaluating the quality of photographs, and subjectively using a perceptual evaluation of synthesised images, face ageing indications, and robustness. They concluded that CycleGAN performed better than AttentionGAN as a whole.

3 Taxonomy of GAN

GANs, deep generative model (DGM), have gained popularity in the deep learning (DL) forum due to their plus over traditional DGMs due to following reasons (1) GANs outperform DGMs. (2) Any sort of generator network may be trained using the GAN foundation. Variational autoencoder (VAE) cannot generate sharp images. Other DGMs may have generator requirements, such as a Gaussian output layer (3), the specification (size) of the latent variable is unrestricted. These positives have made GANs the gold standard for synthesizing image data. Figure 7 demonstrates the analysis and taxonomy of the existing literature from year 2015–2022.

Few advanced GAN including DCGAN [20] proposed by Radford et al. demonstrate that deep convolutional adversarial networks (DCGANs), which are a class of CNNs that contain specific architectural limitations, are a strong contender for unsupervised learning. They present this class of CNNs by the name of deep convolutional adversarial networks (DCGANs)., Pix2Pix [21] devised by Isola et al. look into conditional adversarial networks as a possible solution. also learn a loss function to train this mapping from input image to

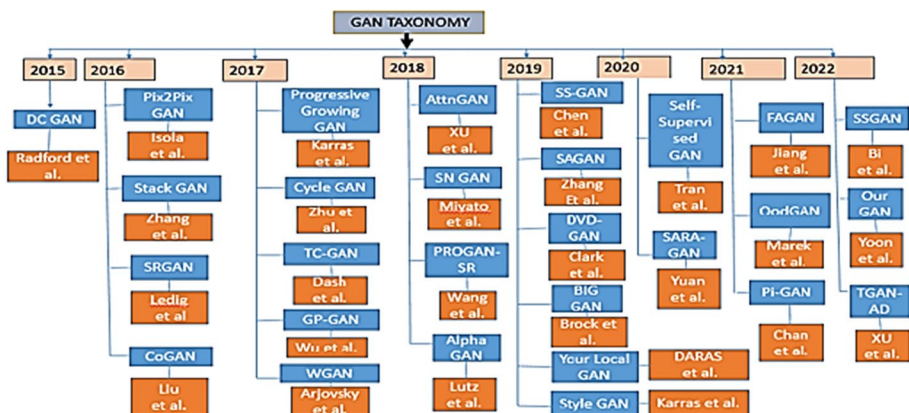


Fig. 7 Taxonomy of the literature of existing GAN models from year 2015–2022

output image. This enables the application of the same broad approach to problems that has previously been used would necessitate radically different loss formulations. StackGAN by Zhang et al. [22] generates 256×256 photo-realistic graphics from text descriptions. Sketch-refinement breaks down the hard problem into smaller ones. Ledig et al. presented a generative adversarial network (GAN) for image super resolution (SR) using a perceptual loss function with adversarial and content losses in SRGAN [23]. The adversarial loss pushes our solution to the natural picture manifold using a discriminator network trained to distinguish super-resolved images from photo-realistic images. Liu et al. suggest CoGAN [24] for learning multi-domain picture distributions. CoGAN can learn a joint distribution without tuples of corresponding images in separate domains, unlike other methods. Karras et al. [25] offer a new approach to training generative adversarial networks called Progressive Growing GAN. Zhu et al. introduced Cycle GAN (CGAN) [26] to learn to translate images from source domain X to target domain Y without paired instances. Our goal is to develop a mapping $G : X \rightarrow Y$ that makes the distribution of images from $G(X)$ indistinguishable from the distribution Y using an adversarial loss. The key is to grow both the generator and discriminator progressively from a low resolution. Dash et al. presented TAC-GAN [27] on the Oxford102 dataset of flowers and evaluated the discriminability of the generated images with Inception-Score and diversity with the Multi-Scale Structural Similarity Index (MS-SSIM). Their inception score is 3.45, 7.8% higher than the previously announced StackGAN, outperforming the state-of-the-art models.

Wu et al. created the Gaussian-Poisson Generative Adversarial Network GP-GAN [28] to combine the strengths of gradient-based approaches with GANs. It's the first GAN study on high-resolution image mixing. They use Gaussian-Poisson Equation to describe the high-resolution picture blending problem, a joint optimization bound by gradient and colour information. Gradient filters give gradient information, inspired by past work. A new study addresses unsupervised learning with WGAN [29] that works on learning a probability distribution and probability density. This is frequently accomplished by creating a parametric family of densities $(P_0)\theta \in \mathcal{R}^d$. AttnGAN [30] by Xu et al. offers attention-driven, multi-stage refining for fine-grained text-to-image creation. The AttnGAN uses a unique attentional generative network to synthesise fine-grained features in different subregions of the picture by paying attention to important words in the natural language description. To stabilise discriminator training, Miyanto et al. developed spectral normalization for SN-GAN [31]. This novel normalizing method is computationally light and straightforward to integrate. Spectral normalization was evaluated on CIFAR10, STL-10, and ILS-VRC2012 datasets. Wang et al. suggest ESRGAN [32] analyses SRGAN's network design, adversarial loss, and perceptual loss and improves these to create an Enhanced SRGAN (ESRGAN). The basic network construction unit is the Residual-in-Residual Dense Block (RRDB) without batch normalization.

AlphaGAN [33] invented by Lutz et al. developed a generator network that predicts appealing alphas using adversarial loss from the discriminator that recognises well-composed photos. Dilated convolutions improve encoder-decoder architectures for convolutional neural networks (CNN) spatial localisation by collecting global context information without downscaling feature maps or sacrificing spatial information. Misc-GAN [34] discovered by Zhou et al. describe the hierarchical distribution of graph structures at different granularities and "transfer" it from the graphs in the region of interest to a unique graph representation. Seven genuine data sets show the framework's efficacy.

SAGAN [35] devised by Zhang et al. found attention-dependent, long-term modeling for picture production. Typical convolutional GANs extract data approximately from lower-resolution feature maps. Clark et al. Dual's Video Discriminator GAN DVD-GAN

[36] a computationally efficient discriminator decomposition scalable to longer and higher-resolution movies. They investigate video synthesis and prediction, obtaining new state-of-the-art Fréchet Inception Distance for Kinetics600 prediction and Inception Score for synthesized using the UCF-101 collection. BIGGAN [37] invented by Brock et al. exhibit recent advances in generative picture modelling by creating high-resolution, varied samples from big datasets like ImageNet that remains challenging. Orthogonal regularization reduces the generator's input variance, allowing a straightforward "trade-off" between sample integrity and variety.

Your Local GAN [38] invented by Daras et al. utilizes two-dimensional geometry and locality are preserved based on sparse attention layer. They demonstrate that substituting SAGAN's thick attention layer in architecture improves FID, Inception score, and pure visuals significantly. ImageNet FID score improves from 18.65 to 15.94. Style-GAN [39] borrowed from the style transfer literature and proposed a different generator architecture for adversarial neural networks with learning. Intuitive, scale-specific control of the synthesis is provided by the innovative architecture, which also results in an unsupervised, automatically learned separation of high-level information (such as stance and identity when trained on human faces) and random variance in the resulting pictures (such as freckles, hair). Self-supervised GAN [40] made by Tran et al. Self-supervised (SS) tasks have been proposed as a solution to the catastrophe fading problem in the GAN discriminator. They find SS task issues that let a generator with a lot of mode collapse work well. SARA-GAN [41] proposed by Yuan et al. developed self-supervised (SS) representation learning utilizing unlabeled data. GAN training has recently used it. SS tasks were suggested to fix GAN discriminator catastrophic forgetting. FAGAN [42] invented by Jiang et al. about the local fusion feature block, extracts visual features at multiple scales using three-pass networks with distinct convolution kernels. The channel attention module, self-attention module, and fusion operation are all parts of the global feature fusion module—enhances the MR image's key features. For training the network, 40 sets of 3D MRI images with 256 slices are used, and 10 sets are employed to measure the proposed methodology.

Ood-GAN [43] discovered by Marek et al. A robust dialogue system must detect OOD utterances. Most dialogue systems achieve this by training on annotated OOD data. Main tasks include intent classification (IC). Pi-GAN [44] by Chen et al. created a revolutionary generative model for high-quality 3D-aware images called Periodic Implicit Generative Adversarial Networks (pi-GAN). Pi-GAN generates sentiment radiance fields by using volumetric rendering and periodic activation functions. 3D-aware picture synthesis using several actual and synthetic datasets yields state-of-the-art results. SSGAN [45] invented by suggest a calligraphic imagery segmentation and classification using autonomous stroking for the evaluation task. Image-to-image conversion is addressed using the strokes segmentation based generative adversarial network (SSGAN) technique. The proposed strategy efficiently produces correct results, unlike existing methods that perform poorly or are unsuitable for practical tasks.

Our-GAN [46] by Yoon et al. present. The first non-repetitive 4K or higher pictures may be generated from a training image using OUR-GAN, the first one-shot ultra-high-resolution (UHR) image synthesis framework. By starting with a low-resolution picture that is visually consistent, OUR-GAN then gradually increases the resolution using super-resolution. TGGAN-AD [47] by propose. The performance of anomaly detection in time series data can be enhanced by using a Transformer-based GAN (TGAN-AD) to extract contextual features of the data. TGAN-discriminator AD's help identify aberrant data. Generators and discriminators calculate anomaly scores. SphereGAN [48] invented by Park et.al offer SphereGAN, an integral probability metric-based GAN. The suggested approach estimates the radial distance on

a hypersphere between two probability distributions, one real and one fictitious. As its objective function is based on a hypersphere, it may determine an half-arc distance upper bound. SphereGAN is stable and converges quickly. We tested three public datasets extensively. GANs' popularity and importance have motivated several reviews [49, 50]. Although GAN training detection tools suffers from some of typical issues [51]. In this way, this review study provides complete coverage of the topic regarding explaining GAN models features, variants, limitations, usages, and challenges in the period 2015 - 2022. It helps researchers to choose a proper architecture and loss function for a specific application under specific circumstances. It shows future researchers how previous researchers dealt with these issues.

3.1 CNN based GAN architecture

Convolutional Neural Network (CNN) deep learning is used in several GAN versions to enhance the model's quality (CNNs). CNNs are specifically designed to work with images, and its basic building block is relatively simple convolutional filters. First, CNN architecture can be utilized to explain visual artefacts created by CNN-based decoders. Second, CNN-based decoders can only output images of low quality (512×512px in the CycleGAN). CycleGAN, which trains generators utilizing an encoder-decoder architecture and cycle reliability. However, these choices are not without challenges. For instance, the deeper the network, the more processing units are required to train it, and in some circumstances, it may not be practicable to centralize all the data required for an effective training process [52]. Collections can have fatal defects when used in real world applications as they may significantly lower the quality of images. The general CNN aligned architecture utilized to construct GAN model can be viewed in Fig. 8.

GAN + CNN coupled models in prediction and classification-based applications are raising concerns that suggests creating combinations of convolutional neural networks (CNNs) as a simple attempt to achieve better results. High-quality image creation makes training unpredictable and time-consuming, yet contemporary high-resolution panels demand graphics with a resolution of 2000 px or above. The various layers of CNN used in GAN architecture are defined in Table 3. To enhance the productivity of the systems (e.g., classification or prediction), machine learning (ML) practitioners often strategically combine or fuse the models called Ensemble of models. In addition to enhancing the reliability

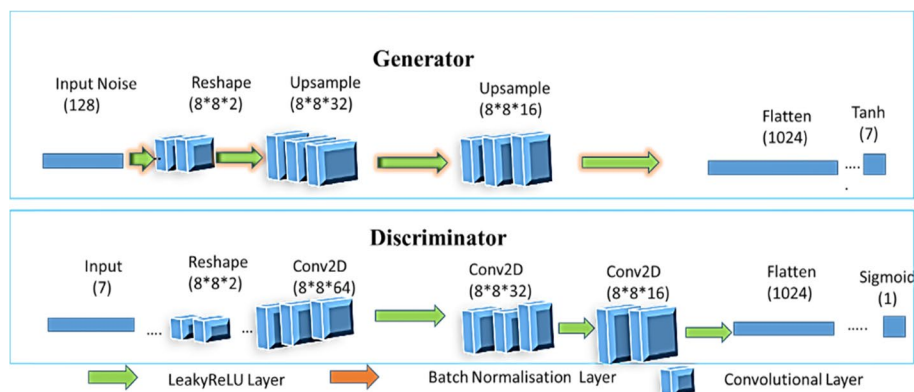


Fig. 8 General CNN based architecture used to establish GAN model

Table 3 Various layers of CNN used in GAN architecture

<i>GAN Layers</i>	<i>Function</i>
<i>Dense Layer</i>	It is used to create Noise for the Generator
<i>Conv2DTranspose</i>	It upscale and convolves the picture at the same time. It's like using UpSampling2D and then Conv2D
<i>Leaky ReLU Layer</i>	It performs better than the ReLU functions because it avoids vanishing of gradient
<i>Batch Normalization</i>	It normalizes the convolution result. This will improve our outcomes
<i>Reshape Layer</i>	It can turn a one-dimensional vector into a three-dimensional array

of a model's conviction, ensemble learning is employed for a wide variety of other tasks, including feature selection, data fusion, incremental learning, non-stationary learning, error correction, and the selection of optimal features. Ensemble techniques in statistics and machine learning combine many learning algorithms to achieve greater prediction performance than each of the component learning algorithms alone. In comparison to a single GAN, an ensemble of GANs can now more correctly reflect the distribution of normal data specially Deepfake identification [53].

3.1.1 Mathematical formulation used in CNN and GAN

- Convolution operation:

$$(I * J)(i, j) = \sum_m \sum_n n(i + m, j + n) \cdot K(m, n)$$

where I is the input image, K is the convolutional kernel, and * denotes the convolution operation.

- Activation function

$$\sigma((I * K) + b)$$

where σ is activation function, K is the convolutional kernel and b is bias term.

- Pooling function

$$Output(i, j) = \max_{m, n} Input(2i + m, 2j + n)$$

Max pooling extracts the highest value inside each pooling window.

- Fully connected layer

$$Output = Activation (Weight \cdot Flatten (Input) + Bias)$$

The input is flattened before the completely linked layer.

- Generator Network

$$G(z; \theta_g)$$

G is the generating function, parameterized by θ_g , and z is the random noise input.

- Discriminator network

$$D(x, \theta_d).$$

The discriminator function, D, is parameterized by θ_d , and the input, x, might be real or generated.

- Objective function GAN

$$GminDmaxV(D, G) = Ex \sim pdata(x)[lodD(x)] + Ez \sim pz(z)[\log(1 - D(G(z)))]$$

where; G stands for Generator, D stands for Discriminator, Pdata(x) = real-world data distribution, P(z) = generator distribution, x = Pdata sample (x), z = a sample taken from P (z), D(x) denotes a discriminator network, G(z) denotes the generator network.

- Training algorithm (Gradient Decent)

$$\theta d = \theta d - \alpha \nabla \theta d J(\theta d)$$

$$\theta g = \theta g - \alpha \nabla \theta g J(\theta g)$$

where α represents the learning rate and ∇ denotes the gradient

Finally, we may conclude that the adversarial training process in GANs consists of a dynamic interaction between the generator and discriminator, led by their respective objectives' functions. The objective functions illustrate the adversarial training process, which is critical for GANs, while the gradient descent updates demonstrate the optimization steps taken during training. Convolutional Neural Networks (CNNs) play an important role in the success of generative adversarial networks (GANs), particularly in picture production and manipulation. CNNs' innate capacity to record spatial hierarchies and local patterns aids in the development of realistic pictures by the GAN generator. Furthermore, CNNs used in the discriminator excel in feature extraction, allowing the network to detect fine characteristics and differentiate between actual and manufactured samples. This ability to extract features and learn representations enables GANs to tackle a wide range of image-related tasks, from style transfer to data augmentation, demonstrating CNNs' adaptability and usefulness in upgrading GAN architectures. The synergy between CNNs and GANs highlights their overall significance in developing the field of computer vision and image synthesis.

In addition to these approaches, Instance normalizing (InstanceNorm) is especially beneficial for activities such as style transfer, where normalizing is conducted separately to each instance or sample. On the regularization front, Elastic Net combines L1 and L2 penalties, providing a versatile technique capable of handling both feature selection and weight reduction. The normalization and regularization approaches used are determined by the dataset's particular properties, neural network design, and the nature of the job at hand. Experimentation and careful selection of these strategies are critical for optimizing model performance and assuring the durability and generalizability of neural network models across several applications.

3.2 Regularization and optimization techniques

Regularization or Normalization techniques are basic approaches used in neural network designs to increase training stability, avoid overfitting, and improve model generalization. In terms of normalization, Batch Normalization (BatchNorm) is extensively used to normalize each layer's inputs by computing the mean and variance across mini batches, reducing internal covariate shift and quickening convergence. Layer Normalization (LayerNorm) broadens this method by normalizing across all characteristics individually for each training sample, which has advantages in circumstances with changing batch sizes.

Group Normalization (GroupNorm) broadens these approaches by normalizing within small groups, making it useful when batch sizes are restricted. Regularization techniques, such as Dropout, are important for avoiding overfitting. During training, dropout randomly turns a proportion of input units to zero, pushing the network to rely on a variety of paths and limiting cell coadaptation. L1 and L2 regularization include penalty terms into the loss function depending on weight magnitudes, discouraging big weights and encouraging sparsity. These strategies work together to increase robust training and generalization, both of which are critical components of developing effective and dependable neural network models.

Optimization techniques are critical in neural network training because they determine how model parameters should be updated to minimize the loss function and improve prediction performance. The cornerstone is Stochastic Gradient Descent (SGD), which iteratively updates parameters in the opposite direction as the gradient of the loss function. Adam, a popular method, combines the advantages of momentum and RMSProp, providing customizable learning rates for specific parameters and demonstrating efficacy across a broad variety of situations. RMSProp modifies learning rates depending on the size of current gradients, whereas Adagrad adapts rates using historical gradient information, which is especially useful for sparse data. AdaDelta enhances RMSProp by addressing sensitivity to initial learning rates. Nadam blends Nesterov's accelerated gradient with Adam's adaptable learning rates. Furthermore, SGD with Momentum includes a momentum factor that aids convergence by dampening oscillations. The best approach relies on dataset features, problem complexity, and network design, and achieving peak performance frequently necessitates empirical testing and adjustment.

The efficiency of various optimization strategies varies depending on the circumstance. Adam, for example, is preferred for its variable learning rates, which make it appropriate for a wide range of applications, whereas SGD with Momentum shines in scenarios with noisy gradients. Choosing the correct optimization method requires striking a careful balance between computing efficiency, convergence speed, and data adaptability. As neural network topologies and datasets get increasingly complicated, the quest for innovative optimization algorithms remains an important field of study to increase training efficiency and model generalization.

In brief, regularization approaches such as Dropout and L1/L2 regularization minimize overfitting by applying penalties to the loss function, hence increasing generalization in neural networks. Stochastic Gradient Descent (SGD), Adam, and RMSProp are optimization algorithms that direct iterative model parameter adjustments to minimize the loss function, resulting in efficient and effective training. These integrated techniques are critical for improving neural network stability, convergence, and generalization over a wide range of tasks and data sets.

4 Popular GAN variants

The existing GAN variants find their importance in distinct areas that are frequently discussed in academic articles. Such endeavors are made to provide established resources of GAN's effectiveness to the growing application areas where GANs are being accepted and tried currently. This review too, has been worked to serve the same purpose and accordingly provides scrutiny of articles that talk about new and traditional GAN variants. Several GAN variants are developed as enhancements of the base GAN model to eliminate

its limitations of instability and complexities at the time of model training. The variants developed and are evaluated many times to identify the prime potentialities and scope of further improvements.

The state-of-the-art GANs, which included base GAN, WGAN, Semi-GAN, C-GAN, LS-GAN, BiGAN, AC-GAN, InfoGAN, SeqGAN, and BEGAN, were reviewed and summarized by Gou et al. [15] concluding that GANs have a lot of potential for virtual-real interaction and integration in parallel systems research. Their experiment demonstrated that GANs provide significant algorithmic assistance for parallel intelligence.

This section presents the works that are most related to different architectures listed as Pix2pix, Cycle-GAN, DC-GAN, BIGGAN, PGGAN etc. of GAN models are analyzed to provide background studies on the framework design and specifics of the GAN model. Figure 9 lists a few popular GANs based on architectures and Loss function. Table 4 provides the various GAN Variants showcasing their merits and improved areas.

4.1 Pix2Pix

Pix2pix is an image processing technology that learns mapping from an input data to a feature map using a conditional generative adversarial network (GCN). It is used for the purpose of converting pictures into images. The discriminator is trained by first producing an output picture from the generator. This is done so that it may be evaluated. The discriminator investigates the relationships between the input, target, and output in order to form an informed opinion on the degree to which the pairs seem to be realistic. The discriminator's weights are then revised in conformity with both the standard errors of the input/output pair and the input/target pair. Further alterations to the weights of the generator are informed by the output of the discriminator as well as the variation between the output and the target image.

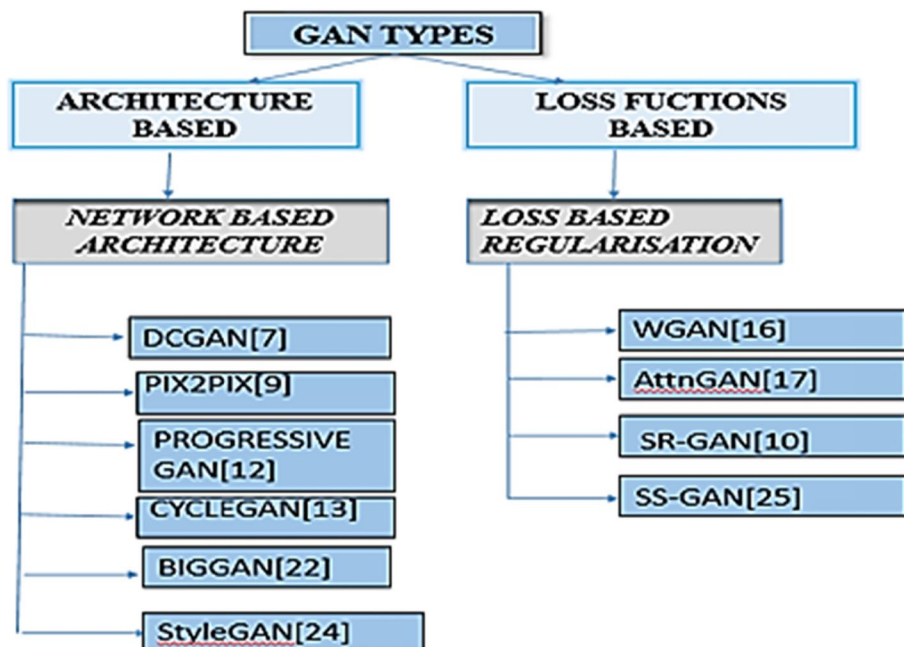


Fig. 9 Few popular GANs on base of architectures and Loss function

Table 4 Various GAN Variants with Key Objective Function

GAN-Variants	Merits	Improvement Area
DCGAN	Easy Convergence and more stable	Need adjustment of training parameters for different datasets
WGAN	Stable training solved theoretically Gradient disappearance Issue	Inappropriate Pruning weights
Pix-2-Pix GAN	Reduction of parameters on large scale	One-to-one paired images in dataset required
Cycle GAN	Convert two Image efficiently styles	Quality of generated target images need improvement
PGGAN	assessment Progress in visual and qualitative	Suffers from inpainting issues
Stack GAN	By phasing generation image sharpness finally improved	Generation involves two stage tasks whole task is halted if one task failed
TP-GAN	Synthetic images involve front face view is well-preserved	Details values are different of generated faces from real faces
SAGAN	Speed up and stable training	Attention extended to larger convolution

The generator is composed of two levels, which are as follows: The encoder, which takes the input picture and converts it into a more compact representation of the features, and the decoder, which mimics the appearance of a conventional generator but is really made up of a series of transpose-convolution layers that do the opposite of what the encoder layers do. The discriminator will not categorize a single picture as either genuine or fake; rather, it will evaluate pairs of images (an input visual and an unidentified visual that is either a target visual or a made visual) and give a real or fake label to each pair of photos. The loss function is

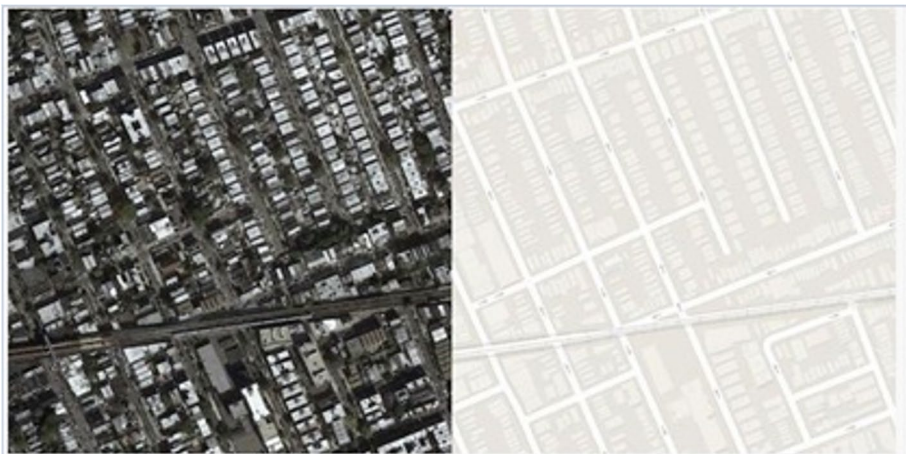


Fig. 10 One image to another image translation from satellite to google map image generated using Pix2Pix using Maps dataset [54]

denoted by the expression $\min G \max E_{x,y} [\log D(x, G(x))] + E_{x,y} [\log(1-D(G(x), y))]$, and its definition can be found here. The problem with pix2pix is that it needs the two picture spaces to be pre-formatted into a single X/Y image that comprises both of the pictures that are strongly connected with each other. Using Pix2pix GAN, satellite photos are translated from one image to another and then converted to Google maps format shown in Fig. 10.

4.2 Cycle-GAN

CycleGAN translates images. It transforms X-domain images into Y-domain images at the generator. Unsupervised learning implies the training pictures don't have labels. Domains don't need image correlation. Two generating networks G and F and two discriminator networks make up Cycle GAN. The discriminator's D_X and D_Y are convolutional neural networks that train tracing: $F: Y \rightarrow X$ and $G: X \rightarrow Y$ to classify an input picture as genuine or fake. Figure 11 shows the translation of horse images into Zebra images using Cycle GAN. In the course of training, two distinct types of training observations are provided as input.

1. One group of observations has paired pictures labelled " X_i, Y_i ," where each X_i is accompanied by its corresponding Y_i .
2. The additional set of observations consists of two different sets of photographs: one class of images from X and another class of images of Y, but there is no match between the two sets. The generators include an encoder, a convolutional network that decreases an image's size, and a decoder, a transpose-convolutional network that alters the image. Using cross-entropy loss, the discriminator determines whether a picture is false or genuine. Discriminator losses are mean squared errors between an image's output and the goal value, 0 or 1, depending on whether it should be classified as fake or genuine. If the image is false, the discriminator's output will have a mean squared error of 1, but if it's true, it will be 0.

4.3 DC-GAN

DCGAN use Unsupervised Representation Learning and is now one of the most widely used and successful architectures for GAN networks. Most of it is composed of convolution layers that do not have any max pooling or layers that are totally coupled together.

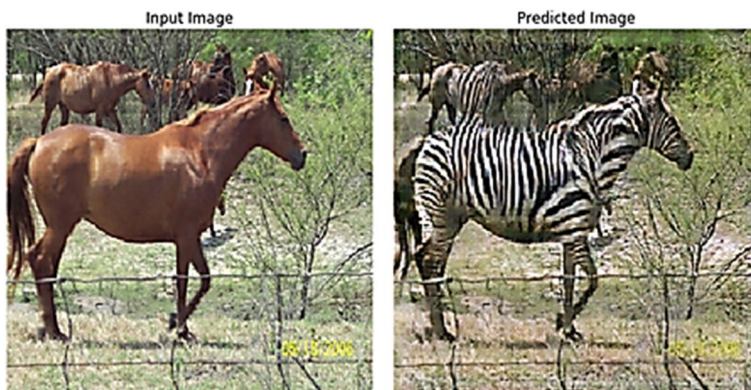


Fig. 11 Translation of images from horse to zebra image generated using CycleGan [55]

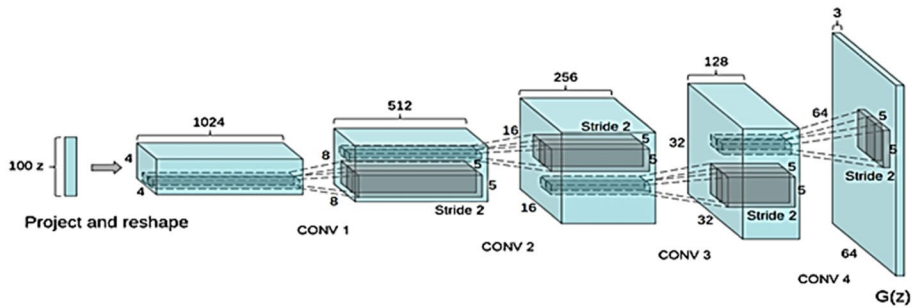


Fig. 12 DCGAN generator's network architecture by Radford et al. [15]

Down sampling and up sampling are both accomplished by the utilization of a convolutional stride and a transposed convolution respectively both the discriminator and the generator in DCGAN employ neural network as shown in Fig. 12. Convolutional-transpose layers come well after convolutional layers. The architecture of the generator's network is depicted in the figure below, which was proposed by Radford et al. for his research on using deep convolutional adversarial generative networks for unsupervised representation learning.

Key Feature:

- Instead than using all-maximum pooling, one should use convolutional stride.
- Use transposed convolution for up sampling.
- Extricate completely connected layers.
- Employ batch normalisation in all layers excluding the discriminator's input and output layers.
- To avoid using tanh in the output, you should use ReLU in the generator.
- The discriminator should make advantage of LeakyReLU.

4.4 big gan

This GAN has the capacity to construct visuals with a superior level of precision and resolution. The BigGAN approach is shown in Fig. 13 which assembles a collection of existing best practices as a way of scaling them up while training class-conditional images and increasing the batch size and number of model parameters. As a direct consequence of this,



Fig. 13 High-Conditional Images of the Best Quality created using BigGAN [32]

high-resolution (big) and high-quality (high-fidelity) photographs may now be generated with relative ease.

4.5 PG-GAN

It is defined as Progressive growing GAN. The network is expanded gradually in GANs are being gradually grown for decent quality, robustness, and variance. To train GAN models to create massive, slightly elevated visuals, GAN is indeed a valid technique. During training, the size of the model is increased in small steps. These can make photorealistic, high-resolution images of fake faces and objects that look incredibly real as mentioned in Fig. 14.

4.6 WGAN and WGAN-GP

To combat issues with GAN programming such mode collapse, which occurs when the generator consistently delivers the very same visuals or a small subset of potential visuals, researcher designed the (Wasserstein GAN) and the (WGAN-GP) (of the training images). An improvement on WGAN is WGAN-GP that replaces weight clipping with a gradient penalty to maintain training stability.

4.7 Style GAN and style 2

A Style-Based Generator Framework for Adversarial Networks," which employs the advancement of ProGAN in combination with image style transfer and adaptive instance normalization (AdaIN), allows for control over the style of the pictures that are created. A Mapping network, which maps the input to an intermediate Latent space, is the underpinning of the Style GAN architecture. After each layer, the intermediate is then processed by the AdaIN, and there are around 18 convolutional layers total. Style 2 GAN has various enhancements over the original StyleGAN, including normalization, gradually increasing and regularization approaches, etc.



Fig. 14 To generate highly realistic faces using Progressive Growing GAN [34]

4.8 SR-GAN (Super Resolution GAN)

The primary objective of this particular category of GAN is to improve the level of detail in an image that has a lower resolution. This is one of the challenges in computer vision that has received the greatest attention from researchers. Both Discriminator and Generator networks make use of Convolutional layers. However, whereas the activation function of the Generator network is the Parametric ReLU, the activation function of the Discriminator network is the Leaky ReLU. The Generator takes the data that is inputted and calculates the average of all the different pixel wise solutions that are feasible.

4.9 SS-GAN (Self Supervised GAN)

Unsupervised representation learning is successfully accomplished using self-supervised (SS) learning. It has been put to use in GAN training. To prevent GAN discriminator catastrophic forgetting, SS tasks were created. This paper analyses how SS tasks affect generator learning. From the analysis, we uncover SS task difficulties that a mode-collapsed generator should be allowed to shine. It provides fresh SS challenges built around a minimax game with many classes. The game's SS challenges drive the generator to understand data distribution and create various samples.

4.10 AttnGAN (Attention GAN)

For the construction of text-to-image conversions, AttnGAN enables attention-driven, multi-stage refinement. By focusing on key natural language words, the AttnGAN can synthesize fine-grained information at various image sub regions. For the purpose of refining the generator's image-text matching loss, it suggests a deep attentional multimodal similarity model.

5 Limitations of GAN systems

Numerous GAN strategies have been created in order to overcome the primary issues with GAN training techniques like gradient vanishing, convergence and mode collapse. As stated before, GAN models suffer from some difficulties that make it unfeasible of use or ineffective as per the desired requirements. Efforts are consistently applied to correct these shortfalls. Diminishing gradients slow or terminate learning, and mode collapse occurred when a generator offers diverse inputs to a single control output class, producing non-diversified examples. Divergent, repetitive, and unstable generator and discriminator training contribute to non-convergence. Some latest articles emphasizing these GAN issues stating as diminishing gradients, which stop or slow down learning, mode collapse, which happens when the generator creates highly similar instances by assigning various inputs to the same class at the output. Non-convergence is also caused by training the generator and discriminator in ways that are different, repetitive, and changeable. All these issues are discussed one by one in details.

6 Convergence failure

One of the most prevalent problems with one drawback of GANs is that they often cannot settle on a solution. During GAN training, it is noticed that theoretically it is guaranteed to converge with direct alteration of the density functions; however, in practice we adjust the sample generation function G . It leads to fluctuations as a consequence. At this point in the process, the model may be trained for an extended period of time, which can provide a wide variety of categories of samples including blurry and noisy pictures. The most extreme kind of non-convergence is referred to as mode collapse. In contrast to prepare additional models, such an image classifier, for use, D and G training losses or accuracy only measure D and G separately, not the GAN's overall performance or the quality of the visuals obtained. The GAN model "works very well indeed" in case the generator and discriminator reach equilibrium if discriminator loss is 0.5. To enhance GAN convergence, other regularization approaches are also applied. This was accomplished by introducing noise to discriminator inputs, as mentioned by Arjovsky et al. [29], and by penalizing discriminator weights, as recommended by Roth et al. [56]. The response from the discriminator D is relatively close to zero since it has been trained to its maximum potential, which causes the convergence rate to decrease. The intensity of the generator is erroneous when the discriminator is not properly trained. The discussion is also demonstrated in the Fig. 15 for visualization of results.

7 Vanishing gradient

When using gradient descent to train artificial neural networks for either generation or classification, the vanishing gradient problem emerges [57]. To compute discriminator and generator losses, gradients, and custom training loop updates. The generator improves with discriminator feedback. The generator will cease training if the discriminator is too strong. Because it can detect false images [58]. The parameters of neural networks are determined utilizing back-propagation methods. These techniques were using a chain rule that moves layer by layer to estimate the gain of the loss function with respect to

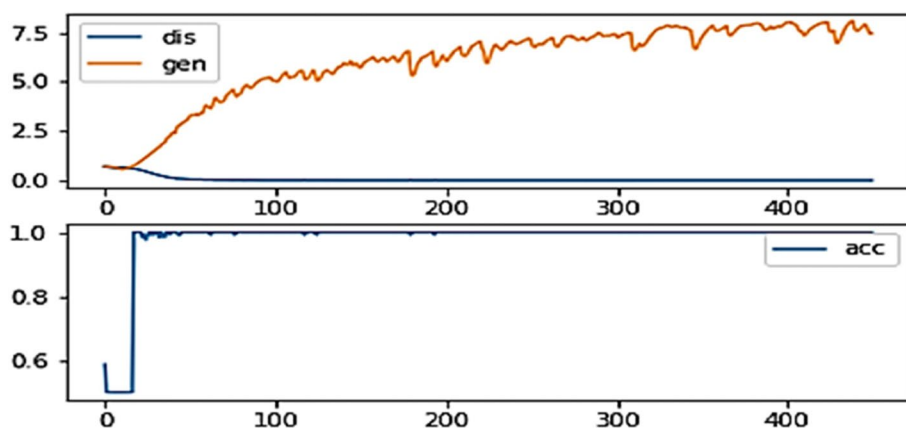


Fig. 15 Loss and accuracy line plots for a generative adversarial network with convergence failure owing to aggressive optimization

the network parameters. The chain rule starts at the final layer and moves backwards until it reaches the initial layer. This allows the derivatives of the initial layers to be determined.

8 Mode-collapse

When the generator creates the same visuals or a miniscule proportion of the training visuals, mode-collapse arises. A successful generator should produce many training-image-like images in all categories. When the discriminator cannot identify fake images, mode collapse occurs and the generator continues to create identical images to deceive it. So, Mode-collapse is a frequent reason for failure when a generator unit only delivers specified categories of actual samples and has little variety among the data. As a result, increasing the variety of the produced samples remains a difficult task for researchers. This is a typical issue with GANs. This problem restricts the GAN's use in most computer vision applications. As illustrated in Fig. 16, Brownlee [59] produced pictures with couple different figure-eight variations, the first one tilting to the left., the second type leaning right, and the third type sitting up with a blur.

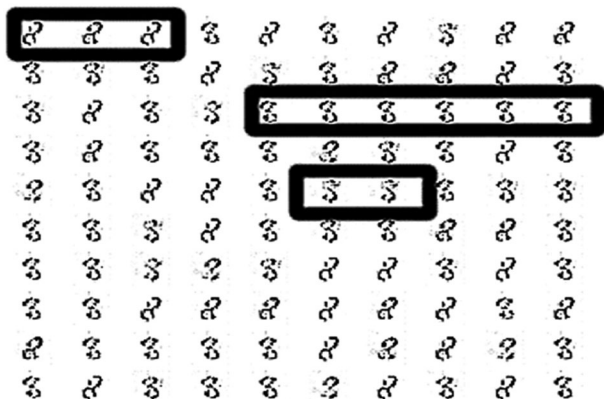
8.1 Instability

Due to the fact that the discriminator often converges more quickly than the generator, it is challenging to strike a balance between the two. Maintaining D and G balance is difficult. Losses during training vary greatly. D and G may stagnate and never improve. Long-term training doesn't always improve generators. Image quality may also get drop. Long-term training doesn't always improve generators [60]. Image quality may also deteriorate over time. This condition is defined as instability of GAN networks.

Efforts are consistently applied to correct these shortfalls. Some latest articles emphasizing these GAN issues and their solutions are given here.

Mescheder et al. [61] established the necessity of the constraint of perfect continuity; we provide an easy-to-understand yet typical counterexample to show that unregularized GAN training is not necessarily convergent in the more realistic scenario of distributions that are not entirely continuous. Additionally, we go through recent regularization algorithms

Fig. 16 An example of 100 drawings of a handwritten 8 from era 315 were generated by a GAN after it experienced mode collapse [43]



that aim to stabilize GAN training. Using instances noise or zero-centered gradient penalties, GAN training converges, according to our findings. On the other hand, it has been demonstrated that Wasserstein-GANs and WGAN-GP do not necessarily converge to the ideal level with a limited set of discriminator updates per generator changes. We present these findings, which provide a fresh justification for the stability issues with GAN training. Based on our research, “we extend the application of our convergence findings to a wider variety of GANs and show local convergence for simplified gradient penalties even when the generator and data distributions are dispersed over lower dimensional manifolds. We find that these penalties operate beautifully in practice and use them to swiftly create high-resolution generative image models for various datasets”.

Rezaei et al. [62] identified a unique ensemble-GAN, or generative multi-adversarial network to address the issue of misclassification when it comes to the semantic segmentation of visuals of the abdomen. The Ensemble-GAN framework incorporated a single generator and a multi-discriminator variant for the purpose of addressing the class imbalance dilemma. This variation delivered a greater approximation than the approaches that had previously been in existence. During the training process, the ensemble model could integrate the assessments of a range of distinct models as it employed a variety of different initial conditions and losses collected from a huge variety of portions of the learning algorithm. The single-producing network reviewed the input visual as a prerequisite to anticipate a matching semantic segmentation image using feedback from the group of discriminator networks. This feedback was obtained by using the image itself. The model was assessed by training the platform on two open datasets, the Chaos 2019 and the LiTS 2017, each with a different imbalance ratio and imaging modality. On well-known abdominal imaging benchmarks, the suggested Ensemble-GAN architecture outperformed competing approaches in the semantic segmentation of healthcare pictures. The semantic segmentation accuracy of the healthy liver left and right kidneys, spleen, and F1 score was 0.93, 0.96, 0.90, and 0.94, respectively. For segmenting the lesions and liver simultaneously, the total F1 values were 0.83 and 0.94, accordingly.

Larsson et al. [63] contrasted convex-concave min-max settings with optimization techniques that actually converge when training GANs using SGDA and Adam. In particular, the Adam-variants of Optimistic Gradient Descent Ascent (OGDA), the ExtraGradient approach, and these methods are discussed. Their performance, robustness, and stability are assessed and compared using the cutting-edge GAN designs U-Net GAN and StyleGAN 2. The empirical evidence on U-Net GAN strongly suggest that the Adam-variants of OGDA and ExtraGradient, as compared to the most popular optimization schemes in contemporary GAN training, regular SGDA and Adam, are more resilient to different choices of hyperparameter settings and less likely to collapse mid-training. However, employing the Adam-variants of OGDA and ExtraGradient had little effect on the final findings for a well-tuned, well-functioning Adam setup like StyleGAN 2. This suggests that while SGDA and Adam are still needed for reaching state-of-the-art performance, these algorithms are more dependable and user-friendly.

Byun et al. [64] suggested a hybrid GAN technique to address the issue of data imbalance and improve the effectiveness of recommendation systems. To create tabular data with both numbers and categorical values, the test used conditional Wasserstein GAN with gradient penalty. Auxiliary classifier loss was also increased by the team to force the model to precisely collect information from the underrepresented class. The GAN realizing for the demonstrated increased use of the theory of PacGAN with discriminator framework to accommodate m-packed samples as input instead of just a singular input. The mode collapse challenge of the conceptual scheme was alleviated by incorporating PacGAN

framework. The studies were carried out to ensure that the model could be validated twice, primarily on the excellence of the generated data, and secondarily on the efficiency of various recommendation models when leveraging the generated data in comparison to the initial data.

In order to analyse variant and in-variant blur, Khan et al. [65] suggested a multi-scaled GAN trained with legacy images leveraging deep learning techniques. For the suggested GAN, various scaled visuals with associated gradients were combined to form a multi-channel single input. To investigate important aspects in the bottleneck region and solve the vanishing gradient problem, residual- and dense-networks were coupled. Gradient-penalty hybrid content loss function was used to lessen the disparity between the generated and real-world photos. Structure sparsity may result in artefacts since some information may be lost in the output. High-resolution anatomical characteristics were retrieved that aids latent image learning, decompression, and edge training proves beneficial. To conduct qualitative and quantitative evaluations, three separate variant and in-variant blur were employed in data - sets from Lai, GoPro, and Kohler. Comparing the suggested method to various preprocessing strategies for image analysis, experiments revealed that it was effective in lowering blur while maintaining structural qualities. Furthermore, the proposed strategy for huge data analysis was proven to be effective across a few publicly accessible datasets.

9 Optimal solution for limitations

The limitations discussed can be resolved by optimization of parameters used in training by generators and discriminators, by selection of activation function, or by using batch normalizations. All these factors are discussed in details one by one in below mentioned subsequent sub sections.

9.1 Selection of activation function

Vanishing gradients arise when an inappropriate activation function is used. Alternatives to the sigmoid and tanh activation functions include ReLU variations and leaky ReLU functions. The vanishing gradient issue is resolved by using activation functions that are not fully saturated. Furthermore, it will hasten the pace at which the two sets of results converge. The generator model employs the ReLU activation as described by Nair et al. [66], with the exception of the convolutional layers, which makes use of the Tanh activation. It has been analyzed in research that when the model was given a bounded activation, it quickly learned to maximize the training distribution.

9.2 Balance between generator and discriminator

It is common for there to be an imbalance between the generators and the discriminator, that might trigger mode collapse and non-convergence. The obvious solution is to give them more variety in their training, which will prevent them from being over fit. Although the researchers have made very little advancement, very few of them believe that this is not a workable or acceptable solution due to the fact that a good discriminator produces a high-quality result [67]. Because of this, some of the focus has been shifted onto cost functions that have gradients that do not vanish completely.

9.3 Boundary equilibrium

This equilibrium was proposed by GAN (BEGAN) (Berthelot et al., 2017) which identifies the connection between the generator and discriminator losses. BEGAN has a basic design and showed impressive performance. It was demonstrated by BEGAN that it is essential for the performance of GANs to strike a balance between two units D and G. The BEGAN architecture, on the other aspect, does not have a structure that is like that of other GANs, hence its applicability is limited. BEGAN made use of an architecture based on EBGAN (Zhao et al., 2016), which employs as a discriminator, an auto encoder utilizes pixel-wise mean squared loss. The discriminator that is a part of BEGAN has two main goals: the first is to auto encode and reconstruct actual images, and the second is to differentiate created images from real ones. By altering the value of, BEGAN can strike a compromise between these two goals while still maintaining the discriminator loss to generator loss ratio. BEGAN created an optimum hyper parameter in the interval $[0, 1]$ in order to retain the optimum between the generator losses and the discriminator losses and the following formula: $= E [L (G (z))]/E [L (x)]$ where L corresponds to pixel-wise auto encoder loss.

9.4 Batch normalization (BN)

The concept of batch normalization (BN) is crucial for deep CNN models to use in order to avoid mode collapse [68]. It performs the function of a preprocessing step, which is then applied to the hidden layers at intermediate to lessen the change caused by internal covariates. This, in turn, helps to avoid by standardizing each mini batch of data, the mode collapse with the help of mean () and variance ().

Any network activations may be batch normalized. Here, we concentrate on transformations with $z = g(Wu + b)$, where W and b are learned model parameters and g () is a non-linearity like sigmoid or ReLU. This is an affine transformation followed by an elementwise nonlinearity. The discussed model has fully connected and convolutional layers.

9.5 Using wasserstein metrics for calculation of loss function

The study by Arjovsky et al. investigated the idea that we should train the critic until it reaches the best level because EM distance is continuous and can be changed. Since Wasserstein is almost always differentiable, it is thought that the more a critic learns, the more stable the Wasserstein gradient will be. When the discriminator uses JS divergence, it gets more accurate and stable gradients. However, the real gradient is 0 because JS is locally saturated, which makes gradients disappear. The observation is that you can get a reliable Wasserstein gradient by practicing it over and over. During testing, both the GAN discriminator and the WGAN critic are trained to do their best. In this case, there is no gradient information, so the discriminator learns right a means to distinguish between genuine and fraudulent samples. This also shows that the EM distance stays the same and can be told apart when the critic is trained well.

10 Applications of GAN

GAN models are vastly recognized and applied in the areas of visual reconstruction and classification. Currently, their application areas are broadened as a tool to produce visual construction from text/audio messages. Also, GAN models are being tried as a prediction

tool to analyses signals, particularly in power systems and medical treatment processes. Figure 17 shows various areas where GAN finds scope and application. The qualities of GAN that helps it to work in diverse domains includes:

- GANs may be used to enhance data by generating new pictures from the training data-set.
- GANs enhance categorization by training the discriminator on actual and false data.
- GANs increase picture resolution.
- GANs turn semantics into pictures and help comprehend machine visualizations.
- In architectural works also, GAN has emerged as a successful model generation and suggestion utility.

11 Improvement in cybersecurity

To stop sensitive information from being leaked and exploited, organizations are using cutting-edge security procedures. However, hackers continue to develop new strategies for obtaining and using user data. Criminal actions like utilizing false photographs and videos to ruin people's reputations or extorting users to keep their information hidden are on the increase and pose a serious threat. Generative adversarial networks have the possibility of being trained to identify fraudulent activity such as this. They are capable of enhancing the dependability of models that are trained via deep learning [69]. Hackers could add harmful information to images, and the neural network can be taught to find it. To train the neural network, researchers and analysts purposefully create fictitious instances. The network becomes more adept at what it does as it examines more photos. To examine and identify harmful encodings that shouldn't be present in the visuals, the Secure Steganography based on generative adversarial network approach was implemented.

12 Generation of animation models

The video game industry stands to gain a substantial amount from the implementation of generative adversarial networks. Because of GANs, the amount of work done by software developers and designers will be reduced. The use of GANs allows for the automatic generation of 3D models, which are necessary for the creation of animated films, cartoons, and video games. The existing dataset of two-dimensional images becomes the foundation for the development of new three-dimensional models by the network. Within a relatively

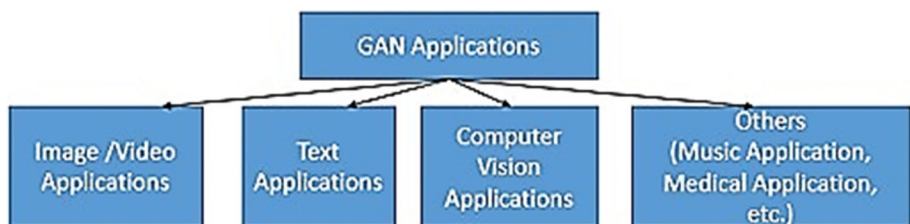


Fig. 17 Classification of GAN applications into various domains

brief span of time, the neural network can analyze 2D photographs and recreating the corresponding 3D models. For example, if we fed 2D images of common household items like tables, chairs, cars, and guns, a neural network could generate a 3D version of the object.

13 Image synthesis and computer vision

Processing images and videos with GAN is among its most well-known usage, including image super resolution, image synthesis, and manipulation. It is one of the most adaptable models for image synthesis thanks to the latter variants of the model, which could use the same theory to generate any data sample based on the distribution inputted to it. This is what specialised variants like CGAN, WGAN, WGAN-GP [70], LAPGAN [71], and InfoGAN specialise in doing. The SRGAN framework was the first to use a GAN model for SR and infer photo-realistic natural pictures for 4 upscaling factors. Three crucial SRGAN components were thoroughly researched by Wang et al., who then enhanced each one to create a better SRGAN with better visual quality named as ESRGAN [32]. The notion of using proportional authenticity rather than absolute number to predict discriminator efficacy is borrowed from relativistic GANs [72], and is used in ESRGAN.

14 Improvement in healthcare industry

One of the industries that stands to gain the most from AI, neural networks, and generative adversarial networks is the healthcare and pharmaceutical industry. GANs can be used to find tumors in the body. By comparing images of tumors to a series of photos depicting functional organs, the neural network can be used to find tumors. The neural net can address risks in visuals and scans of the patient checking for differences between them and the images in the dataset. When generative adversarial networks are used, cancerous tumors can be found faster and more accurately. It saves money for both the patient and the doctor. But the most important thing is that generative adversarial networks could help save people's lives.

Drug discovery is another area of health care where generative adversarial networks can help. The networks can be utilized for generating the organic structures of medicines that can further be harnessed to target diseases and treat them.

15 Image to image translation

The process of extracting data from images can be accomplished through the assistance of GANs. It works out for a variety of translation tasks, including translations from text to images, from images to images, and from images to images semantically. Translations of images into other images: GANs have the potential to be utilized in image-to-image translations that will be utilized for varied translation operations, including:

- a. The conversion of satellite/aerial imagery to Google Earth or Google maps.
- b. Changing specifics day to darkness and back again in the editing process.
- c. The process of coloring photographs that were originally taken in black and white.
- d. Transforming sketches into photographs in color.

CycleGAN, TripleGAN [73], and DiscoGAN [74] are the most popular versions utilised to accomplish this goal. Zhu et al. featured CycleGAN and further neural machine translation examples, including converting pictures to artworks and a horse from a zebra., and summer to winter photos. Wang et al. outlined a technique for semantic image-to-photo translations using conditional GANs. To generate a realistic visual from a semantic sketch that is provided as input, conditional GANs [75] can be utilized. Some examples of things includes, with the use of the given semantic information, photorealistic translations of a metropolis, residences, a real face, natural surroundings, and automobiles may be discovered.

16 Text/Audio to image generation

The neural net can synthesize visuals autonomously from texts with GAN. TAC-GAN advented by Dash et al. in year 2017 synthesizes images from text descriptions. It is build on AC-GAN by subjecting produced visuals to a text description, Gorti et al. [76] worked on cycle Consistent Adversarial Networks devised for Text for Image for Text Translation. They recommended an annotating network to caption on manufactured visuals, and they made leverage of the distinction between generated labels and ground truth captioning to further enhance the network. Reed et al. [77] Generating a new to link these advancement in text and picture modelling and transforming visual notions from letters to pixels, deep architecture and GAN formulation are used. He demonstrated how to utilize a model to create lifelike pics of birds and flowers extracted from extensive detailed descriptions.

Chang et al. [78] suggested “Feature-Aware Generative Adversarial Network (FA-GAN)” to combine a discriminator that could really assess its own productivity and a functionality dropout to create a high-quality visual. In order for the discriminator to recover good indication, a self-supervised discriminator with an auxiliary processor was first created in the study. In order to give the generator more direct supervision, a feature-aware loss that makes use of the visualization of the characteristics supplied by the self-supervised discriminator was also implemented. In accordance with the findings of the experiments undertaken on the MS-COCO collection, the suggested methodology resulted in a significant increase to the state-of-the-art FID score, which dropped from 28.92 to 24.58.

Novel convolutional recurrent GAN (CRGAN) architectures for voice improvement were proposed by Li et al. In order to allow for comparing the differences with other GAN-based systems, multiple loss functions were implemented. Recurrent layer advantages were also investigated. The outcomes of the studies substantiated the appropriateness of the postulated CRGAN model outperformed other non-GAN based systems and the SOTA GAN-based models using the identical loss functions, demonstrating the advantages of using a GAN for voice augmentation.

To create images from vocal communications, Zhang et al. [79] used a GAN technique that included spectral norm, projection discriminator, and auxiliary classifier as opposed to naïve conditional GAN. Both subjectively and objectively, the model was able to produce photographs of higher quality. Nearly three-fourths of those who provided comments on their suggested model concurred that it might produce visuals associated with sounds. The model demonstrated that it understood the association between sounds and visuals to some extent by responding to inputs of the same sound at various levels with varied scales of change.

Without using interpretation, Wan et al. [80] sought to convert speech signals into visual cues. In order to improve generalisation on new classes, a voice encoder was created to represent the input speech signals as an embedding feature. It was then trained utilising teacher-student learning using an image encoder that had already been trained. A stacked generative adversarial model was then employed to create excellent images based on the insertion attribute. The effectiveness of our suggested method to convert the raw speech signals into images without the intermediary text representation was demonstrated by experimental results on both synthetic and actual data. Ablation research provided more details regarding the method under investigation.

16.1 Visual reconstruction and classification

GAN as a deep learning approach provide an acceptable solution to the low-resolution to high-resolution picture translation issue. For instance, it can build 7T from 3T MRI and magnetic resonance imaging (MRI) from computed tomography (CT) pictures, allowing for the creation of multimodal datasets from a single modality.

A multi-GAN [81] and multi-application study was carried out by Skandarani et al. [82] to evaluate the improvements that GANs potentially contribute to the discipline of medical image recognition. They performed experiments on three modalities of medical imaging and organs using diversity of GAN designs, running from the simplest and foundational DCGAN to the most comprehensive style-based GANs. “cardiac cine-MRI, liver CT and RGB retina images”. To gauge the visual acuity of the images produced by GANs, their FID score was estimated from training data taken from well-known and frequently used datasets. By evaluating the U-Net trained on these produced photos’ segmentation accuracy, their experiment further evaluated the efficacy of the methodologies. The findings showed that not all GANs were created equal, with some being far better suited to applications in medical imaging than others. By FID standards, the best GANs were able to produce realistic-looking medical images that might deceive qualified competent in the vision Computer simulation and determined to adhere with specified constraints. No GAN, however, appeared to be able to replicate the complete richness of a medical dataset, according to classification performance.

16.2 Architectural design

Newton et al. [83] made a novel contribution regarding the use of GANs for the generation and study of architectural plans in relation to the work of particular architects. In this experiment, GANs were trained properly to synthesize Le Corbusier’s structural drawings, and they were then employed to offer analytical insight. The trials showed how effective various augmentation methods are for usage by constructors when dealing with tiny datasets.

Aditi et al. [84] concentrated on the development of volumetric designs based on a supplied programed graph. The researchers suggested a novel 3D model called voxel graph that was both compact and expressive for creating geometries in place of producing dense 3D voxels. The pipeline was trained using an adversarial framework, and their generator was a cross-modal graph neural network that connected the input programed graph and the output voxel graph via a pointer mechanism. A user study was used to evaluate the created designs qualitatively, and three metrics—quality, diversity, and connectivity accuracy—were used to evaluate them quantitatively. The researchers demonstrated that the

established model produced 3D volumetric designs that were realistic and performed better than baselines and earlier techniques.

16.3 Visual prediction of signals

Using machine learning techniques, Li et al. [85] offered a preliminary solution for the use case of monitoring the surveillance system in which sporadic unexpected events take place. The method was designed to prevent the need for human resource management to investigate any suspicious activity captured on surveillance system recordings. They developed a model that used past and present video frames are used to identify anomalous behavior and predict future events. To distinguish an anomalous or normal dynamic visual frame, they constructed a reconstruction-based One-Class Classifiers (OCC) using GAN architecture. Only normal data were employed to train the prediction model's one generator and two discriminators, atypical or aberrant samples are regarded as outliers. The model may thus be predicted to recreate results poorly when it received an abnormal sample (an outlier), aiding in the detection of the faulty frame.

Li et al. [86] proposes TTS-GAN, a transformer-based GAN that could efficiently produce genuine time-series data patterns, of any duration, that are blended to seem like the genuine product, was introduced by [38]. The GAN model's discriminator and generator networks were both constructed utilizing a pure transformer encoder structure. The closeness of real and produced time-series data was illustrated using visualizations and dimensionality reduction techniques. The investigation also contrasted the generated data's quality with that of the best available substitute, an RNN-based time-series GAN.

Festag et al. [87] highlighted and summarized research efforts that currently occurring place in the realm of time series prediction and imputation utilizing GANs. Even though GANs started out intended to operate just in the current frame; however, subsequent research has shown that they are also proficient of attributing and estimating non-visual time - series data.

Their research distinguished between forecasting, which defined the estimation of future elements, and restoration, which characterized the putting in of missing values at intermediate levels. It was important to look into how these methods were used specifically in the biomedical field. Regardless of the application domain, specifically, it included research on GANs for time - series data interpolation and prediction. 33 records were ultimately determined to be valid and arranged in accordance with the illustrations of GANs, encompassing their geometries, inefficiencies, sources, and products.

16.4 Text recognition

Vlachostergiou et al. [88] used and assessed Generative Adversarial Network (GAN) methods to validate their efficacy for learning document-, phrase-, and aspect-level representations of natural language. In comparison to the employment of GANs in picture synthesis, the researchers claim that the faculty was mostly disregarded. Utilizing the Movie Review (MR) Dataset, the Fine-grained Sentiment Set of data, and the 20 Newsgroups Corpus, they conducted substantial research to test the application experimentally (FSD). Their experimental research revealed that all three of the aforesaid layers of presentation of natural language texts might be learned by GANs.

The task of handwritten signature detection was studied by Siyue et al. [89] by deploying GAN technology for the very first time. The idea of dual learning as the foundation

for this, a unique network called “SIGAN (Signature Identification GAN, SIGAN)” was proposed. The identification threshold was set at the SIGAN-trained discriminator’s losses value. It was feasible to determine the integrity of the test handwritten signature by comparing the threshold and loss value of the test image acquired by the network. The experimental data set for this study consisted of five hard pen-type signatures, some legitimate and others deliberate impersonations. According to the experimental findings, the SIGAN-based signature identification model had an overall accuracy obtained was 91.2%, which was 3.6% higher than the standard image classification technique’s dependability.

16.5 GAN and parallel vision

Fei-Yue Wang [90, 91] presented scientific frameworks like ACPs (Computational Experimentations, Simultaneous Execution, and Artificial Societies) strategy aiding modelling as well as controlling complex systems in 2004. Systems that run in parallel encourage hybrid online-offline engagement. Computational experiments supports to generate and evaluate computer simulations, virtual systems are constructed to approximate the actual product, and parallel execution improves system performance. The idea of combining the Authentic and the Artificial in parallel systems may be expanded upon by using GANs. As a group of very effective unsupervised learning, GANs have the potential to be included into the academic research of paralleled intelligence. In the realm of addressing complicated technologies, the ACP method finds particular use in the form of parallel control [92, 93]. GANs enable the production of artificial-scene pictures in parallel vision. To train vision models, GANs may create enormous, diverse visual features that could be linked with those other, relatively traditional datasets. As a result, conceptual framework for judging model’s capacity to generalize is enhanced.

16.6 Other fields

16.6.1 (Music applications)

It has been established that GAN and its varieties, such as the C-RNN GAN proposed by Mogren et al. [94], the Object Reinforced GAN (ORGAN) discovered by Guimaraes et al. [95], and the SeqGAN presented by Yu et al. [96] may be useful in the area of music production.

16.6.2 (Medical applications)

The medical industry is one of the most significant applications for generative adversarial networks (GANs), where they are used for tasks such as the synthesis and design of DNA illustrated by Killoran et al. [97], chemGAN proposed by Behenda et al. [98], and medical imaging. Additionally, it is used in dental records [99] and suggestions [100] given by medical professionals. Med-GAN by Choi et al. [101] uses generative adversarial networks to make patient records with more than one label. It can create convincing fake medical files. Using an autoencoder and generative adversarial networks, To synthesize steeply latent variables (such as symmetric and counting features) from legitimate medical data, “medGAN” leverages a deep convolutional based adversarial network.

16.6.3 (Sentimental analysis)

GAN is also utilized in sentiment analysis, which is a key subset of natural language processing that combines computer linguistics and text mining to capture larger emotions from publicly available viewpoints [102]. Negations have an important function in linguistics, particularly in sentiment analysis, since they can influence the polarity of other textual elements. In terms of structural formation, negations can exist in two forms: syntactic negation and morphological negation.

17 Evaluation metrics

One major benefit of adopting GANs has been the elimination of the need for sophisticated Markov chain sampling and inference. In some ways, GANs have alleviated the burden of more meticulously training a model. Using GANs also increased the productivity of the individual models. But, GAN with new variants requires need of new parameters to deal with performance issues. The advancement of evaluation metrics of GAN can enable in minimization the various consequences of the pertaining GAN issues. GAN models are basically measured qualitatively and quantitatively.

Non-numerical qualitative measures involve subjective human evaluation or comparison. Here are five qualitative GAN generator model evaluation techniques. Evaluating Mode Collapse, Rating and Preference, Mode Drop, Nearest Neighbors, and Network Internals. Quantitative GAN generator assessment calculates accurate image quality ratings. There are 24 quantitative methods for evaluating GAN generator models. FID, Recall, F1 score, Precision, Inception Score (IS), Geometry Score, Discriminator Rejection sampling, Annealed Importance Sampling, Tournament Based Metrics, etc. Accuracy refers to the percentage of correctly classified test data, whereas F-measure evaluates a class based on the harmonic mean of precision and recall [103]. By using adversarial strategies, GAN is able to succeed where other models have failed like Boltzmann Machine [104]: efficiently fitting real-world data distributions into an output function. A successful GAN model should have acceptable picture quality (not fuzzy) and diversity (images that approach the rehearsal set of data's distribution). To test the GAN model, visually analyze the trained or inferred pictures. Table 5 shows comparative study of various GAN models. Table 6 presents scores for parameters IS or FID considered important factors for improving its performance. Merits and demerits of GAN Variants with focus on areas of improvements for making a better GAN are analyzed in Table 6. When evaluating GAN performance, both the quality of the generated samples and the convergence of the training process are considered. Here are some essential performance measures for GANs, along with their mathematical basis:

1. Precision, Recall and F1-Score

These metrics assess the discriminator's ability to discriminate between actual and produced samples. Precision is defined as the ratio of properly recognized produced samples, recall as the ratio of correctly identified real samples, and F1 Score as the harmonic mean of precision and recall.

2. Fréchet Chem Net Distance (FCD).

Table 5 Comparative study of scores of parameters IS and FID of existing GAN Variants

GAN Variant	Dataset	Inspection Score (IS)	Fréchet Inspection Distance (FID)
FC-GAN	CIFAR10	6.41	42.6
Sphere GAN	ImageNet	166.5	74
SN-GAN	ImageNet (350 K images)	36.8	27.621
BIG GAN	JFT-300 M	50.88	1.94
SA-GAN	ImageNet	52.52	18.65
LS-GAN	CIFAR10	6.76	29.5
DCGAN	CIFAR-10	6.69	42.5
Sphere GAN	ImageNet	166.5	74
BiG-GAN	CIFAR10	9.22	14.7
Sphere GAN	ImageNet	166.5	74

Table 6 GAN Variants “IS” or “FID” Values

GAN Variant	Dataset	Metric (IS or FID)
Style GAN2	Text2Face GAN	FID:118.097
WGAN-GP	CIFAR-10	FID:40.2
	STL-10	FID:55.1
PG-GAN	CIFAR-10(Unsupervised)	IS:8.80
Attn-GAN	CUB dataset	IS: 14.14
	COCO dataset	IS:170.25
Sphere GAN	LSUN	FID: 16.9
TAC-GAN	Oxford- 102(Flowers dataset)	IS:3.45
SN-GAN	CIFAR-10	FID:25.5
	STL-10	FID:43.2

FCD calculates the distance between the distributions of actual and produced samples using their feature representations. It applies the Fréchet Inception Distance to other fields, such as chemistry.

3. Kernel Inspection Distance (KID)

KID compares the feature representations of actual and produced samples using the Maximum Mean Discrepancy (MMD) between kernelized distributions.

4. Inception Score (IS) is a metric that accurately depicts the photographs’ quality and diversity. The Inception Score evaluates the quality and diversity of produced samples. It use the Inception v3 neural network to calculate the conditional entropy of class predictions for produced data. A higher Inception Score suggests superior quality and diversity.

Formula: $IS = \exp(\mathbb{E}_x [\mathbf{D}_{KL}(\mathbf{p}(y|x)||\mathbf{p}(y))])$ where, $\mathbf{p}(y|x)$ is the class distribution given a produced sample , $\mathbf{p}(y)$ is the marginal class distribution, and $\mathbf{D}_{KL}$ represents the Kullback-Leibler divergence.

5. Fréchet Inception Distance (FID) is a metric that not only evaluates the produced pictures on their own, but also compares the genuine and false versions of an image. It determines the Frechet distance between the feature representations taken from the

Inception v3 network to determine the similarity between the distributions of actual and produced samples.

Both the inception score and the FID score are based on the inception v3 algorithm. In this paradigm, the computer-vision-specific properties of an input picture are captured in the coding layer (the final pooling layer before the output categorization of images). All the activations are computed for a database of both actual and artificial photos. To summarise the activations as a multivariate Gaussian, we determine the mean and covariance of the pictures. The activations in both the actual and synthetic pictures are analysed statistically in this way. Frechet distance, also known as Wasserstein-2 distance, is used to determine how far apart these two distributions are.

The inception score provides an evaluation of the quality of a set of synthetic pictures based on how effectively they are classified as one of 1,000 known items by the state-of-the-art image classification algorithm Inception v3. The scores are a sum of the marginal probability integral and the quality of the conditional class predictions for each synthetic picture (diversity). The purpose of creating the FID score is to compare the statistical characteristics of a set of synthetic photos to those of a set of real images from the target domain. To better capture the similarity of produced pictures to genuine ones, the "Fréchet Inception Distance" (FID) was established for evaluating the effectiveness of GANs at image production better than IS(Inspection Score).

18 Datasets for different GAN models

The datasets used for training and testing Generative Adversarial Networks (GANs) have a major influence on their performance and generalization. Here are some frequently used datasets in the context of GANs.

- **MNIST:**

Description: A series of 28×28 grayscale photos of handwritten numerals (0–9).

Due to its simplicity and modest size, this dataset is frequently used as a benchmark for GANs in practice.

- **CIFAR 10 and CIFAR 100:**

CIFAR-10 has 60,000 32×32 color pictures divided into 10 classes, and CIFAR-100 includes 100 classes.

Use Case: Widely used for GANs that operate with color photos and other complicated visual information.

- **ImageNet:**

A large-scale collection with over a million tagged high-resolution photos from 1,000 distinct categories.

Use Case: Commonly used to train GANs on a wide range of complicated real-world pictures.

- **CelebA:** This dataset includes over 200,000 celebrity photos with annotations on different characteristics.

Use Case: Frequently used for face-related GAN applications, such as facial attribute modification and creation.

- **LSUN (Large Scale Scene Understanding) Dataset:** Description: A collection of scene-centric photos from many categories, including bedrooms, churches, and bridges.
Use Case: Suitable for GANs that generate realistic sceneries and backdrops.
- **DeepFashion:** This dataset has over 800,000 varied fashion photos with thorough annotations.
Use Case: Used in GANs to generate and alter fashion photos.
- **Art Dataset (such as WikiArt):**
Description: Sets of artworks from diverse artists and genres.
Use Case: GANs are used to generate art, replicate certain art styles, or create unique artistic creations.
- **Cityscapes:** Description: A collection of high-resolution urban street pictures with semantic annotations.
Use Case: GANs are used to generate realistic urban environments and improve urban-related applications.
- **Maps collection:** This collection includes maps and satellite photos.
Use Case: GANs can generate maps, improve navigation systems, and simulate realistic geographical settings.
- **Brain MRI Datasets (e.g., ADNI):**—Medical imaging datasets with brain MRI images for Alzheimer’s disease research.
Use Case: GANs are used to generate synthetic medical pictures and supplement existing data.

19 Research findings and discussion

GAN models that are traditionally evolved as image generation application has now grown in its usage areas. Focused on this aspect, this review article has discussed on the academic works on GAN applications where the approach is made more conducive for its usages, the GAN variants, limitations of GAN models and areas where GAN approaches are successfully used. Based on the reviews of current scholarly articles that are published during 2016 – 2022, the following observations are pointed out in the nutshell:

- GAN and Deep learning approaches are integrated currently to expand and enhance the applications of the model. Researches are emphasizing the substantiality of database parameters to effectively train GAN generators and discriminators.
- Robust model evaluation techniques are adopted to optimize the GAN model performances eliminating the chance of defects. Also, the evaluation rules are combined and applied to measure GAN model performance, which are giving better outcome.
- GAN models are currently developed for visual generations of architectural planning, 3D visual model generation, and conversion between image and image/text and image/audio and image related jobs. GAN models are getting more attention in medical image (such as, X-Ray, Body scans, etc.) based diagnosis.
- GAN models are emerging as a powerful tool to predict signals and interpret them visually. Mostly, in medical application and power supply units, GAN prediction applications are used currently.
- GAN tools are proved to be effective as natural language processing applications.
- GAN model shortfalls, such as, training difficulties, model instability with data heterogeneity and data volume, false prediction, noise and poor performances due to data

inaccuracy, etc. are deeply analyzed and GAN variants are evaluated extensively to optimize their performance.

20 Conclusion

Based on the findings mentioned above, we can say that scope and possibilities of GAN applications are growing with the emergence of their new variants and improvements in their frameworks. Many GANs are memory constrained, and because the models are so big, even just training a batch size of one or two will fill the GPU RAM just holding one or two samples. Training a simple neural network requires some computational work, but training a dueling network is far more taxing on the machine, especially in terms of memory. Side-by-side, it is equally important to have the exact evaluation metrics for these applications to indicate their worth and suitability of usages. GANs are currently getting scopes as a predictive tool. It is vastly in use to generate and interpret architectural planning, 3D visual model generation, and conversion between image and image/text and image/audio and image related jobs. Generalization of GAN for handling wild scenarios with more effective evaluation metrics would be the promising area for future research in this field. In nutshell, we are concluded with following research ideas in this field:

- The task dependency issue is crucial in estimating the performance of GANs. Emphasizing the studies are needed to determine how well a certain generative model would operate in the given context.
- Comparative empirical and analytical investigations of existing measures are needed.
- Models are compared across several metrics under controlled settings like architecture, optimization, hyper parameters, computing budget, etc. For rating generative models and identifying their flaws, effective assessment metrics are crucial and require exploring.
- It is not yet known how much training data is memorized by generative models. A Well-known method “use of nearest neighbors” is a frequent method of testing memory which suffer from number of flaws that require to optimize.
- It has been discovered from evidence that implies evaluation parameters like IS or FID as a loss function can enhance the samples of a generator; however, when this happens and when it does not remain unclear and should be the focus of future study.
- More study is needed to fill the gap in faked image analysis caused by the three primary technical obstacles encountered during training of GANs: mode collapse, non-convergence, and instability. The researchers are so inspired to suggest cutting-edge remedies for the fundamental training problems faced by GANs in this field.
- Images produced by generative models have been the primary metric by which they have been judged. The extent to which these concepts may be generalized or used equitably is less well studied and need to explore more.

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Data Availability The data is available on request.

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